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Physics-informed neural networks and variants in weather and hydrological modeling: a systematic review

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Abstract

Physics-informed neural networks (PINNs) and other physics-informed machine-learning techniques have proven to be a powerful way to link data-driven inference to governing equations in weather and hydrological modeling. This systematic review brings together the literature on PINN architectures, variants, application areas, governing equations, data sources, validation methods, and operational limitations. Boolean search strings were used in major scholarly databases, and records were filtered based on PRISMA-guided inclusion and exclusion criteria, along with quality-appraisal criteria. The review separates relatively well-established applications, such as groundwater-flow inversion, shallow-water flow, and Saint-Venant flow modeling, from some sparse-observation reconstruction tasks from early proof-of-concept studies in high-dimensional atmospheric forecasting and digital-twin deployment. The synthesis reveals that PINNs can have a positive impact on physical consistency and decrease the need for dense labels, but they have practical limitations due to training instability, stiffness, sensitivity to loss weighting, computational costs, scalability in high-dimensional domains, uncertainty quantification, and sparse field-scale validation. Finally, the study provides conditions under which the use of PINNs is advantageous over purely data-driven neural networks, surrogate models, and conventional numerical solvers, and highlights avenues for operational use in combination with data assimilation, forecasting systems, and hydrometeorological digital twins.

Keywords: Deep learning; Physics-informed neural networks; weather modeling; hydrology; hybrid modeling

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1. Introduction

The rapid convergence of physics-based modeling and data-driven artificial intelligence (AI) has revolutionized the approach of scientists to complex problems in Earth system science (Waqas et al., 2024; Zhao et al., 2024b). Over the past decade, machine learning (ML) and deep learning (DL) have been used to detect nonlinear patterns and spatiotemporal associations in large environmental datasets, which can be used to improve the predictive accuracy of weather forecasting and hydrological modeling (Mathew et al., 2021; Waqas et al., 2023). However, conventional ML models, despite their flexibility and computational efficiency, often disregard physical constraints, resulting in poor generalization, physical inconsistencies, and reduced reliability in unobserved environmental conditions (Barbierato & Gatti, 2024; Zaker Esteghamati et al., 2025). Physics-informed neural networks (PINNs) have been proposed to fill this gap by integrating physical systems (typically represented as ordinary or partial differential equations (ODEs/PDEs)) into neural network training (Habib et al., 2025; Lawal et al., 2022). PINNs embed physics in the loss function, integrating data-driven learning and first-principles modeling to ensure accurate and physically consistent predictions, even in regimes of data scarcity or noise (Haitsiukevich, 2025).

In weather and hydrological sciences, non-linear dynamics, including atmospheric convection, precipitation, evapotranspiration, and runoff generation, are best modeled by integrating complex multi-scale processes that are solved using the Navier-Stokes, energy balance, and mass conservation equations (Kim, 2013; Sivapalan et al., 2001; Yang et al., 2021). Numerical weather prediction (NWP) and hydrological simulation models are computationally intensive and sensitive to errors in parameterization and boundary conditions, as well as data assimilation uncertainties (Kashinath et al., 2021; Schultz et al., 2021). Conversely, ML/DL models can quickly mimic such processes but tend to overfit observational data, often failing to respect conservation laws (Dhanka et al., 2025). With the introduction of PINNs, a middle ground is offered that leverages both paradigms, imposing physical constraints on the learning process to increase interpretability, stability, and efficiency (Cai et al., 2021; Lawal et al., 2022). The use of PINNs in the atmospheric and hydrological setup, therefore, represents an essential methodological development that can overcome the drawbacks of exclusively physical and exclusively data-driven models (Zeinali et al., 2025).

PINNs have been highly successful in numerous geoscientific disciplines (Zhao et al., 2024b). More specifically, in groundwater and flood modeling, PINNs have been used to estimate solutions to the Shallow Water Equations and have been shown to scale well for reconstructing flow fields using sparse or incomplete observations, while remaining physically faithful (Donnelly et al., 2024; Zhang, 2024). They have been used in atmospheric and climate modeling to simulate the dynamics of turbulent flows, parameterize subgrid processes, and downscale climate projections, while remaining energy- and momentum-conserving (Zhao et al., 2024b). The PINN framework has also been applied to related areas such as power systems (Huang & Wang, 2022), geotechnical engineering (Yuan et al., 2025), and aerospace simulation (Tkachov & Murashko, 2025), demonstrating its high versatility. Although these methods have been used in engineering and physical sciences, their systematic use and assessment in weather and hydrological modeling remain scattered and understudied (Waqas et al., 2024).

Existing reviews of physics-informed ML (PIML) and PINNs have been more subjective than existing reviews, focusing on a more general geoscientific or engineering perspective. Habib et al. (2025) provided a broad overview of the applicability of PINNs in geosciences, including hydrology, seismology, and atmospheric sciences. They mentioned that these could be used to improve computational efficiency and predictive accuracy (Habib et al., 2025). Similarly, Chew et al. (2025) studied the applications of PIML in the design of urban infrastructure and the development of resilience, noting that integrating physical constraints into ML models can enhance generalization and reduce the need for extensive sensor networks (Chew et al., 2025). Yuan et al. (2025) found the interpretability benefits of PIML in the geotechnical context (Yuan et al., 2025), but Huang and Wang (2022) identified several paradigms of PIML, such as physics-guided (Huang & Wang, 2022). However, none of these reviews have addressed the unique issues and methodological challenges of weather and hydrological systems, where multi-scale coupling, stochastic fluctuations, and non-stationarity are prevalent phenomena. Thus, a dedicated synthesis should be conducted to examine the advances made by PINNs in constructing hydrological and atmospheric models.

This review was designed in response to the growing recognition that traditional hydrological and meteorological models struggle to balance physical rigor with data adaptability under varying climatic conditions. Conventional PIMLs, such as SWOT, VIC, and WRF, necessitate considerable calibration and are highly computationally intensive, whereas data-driven models are not robust to extrapolation (Eder, 2024; Schultz et al., 2021; Waqas et al., 2024; Xu et al., 2023; Zhang et al., 2025).

Despite their limitations, PINNs facilitate the implementation of hydrological and atmospheric governing equations within neural network structures (Ali et al., 2024). For example, the solution of the one-dimensional Saint-Venant equation or the two-dimensional Shallow Water Equation using PINNs has resulted in significant computational cost savings compared to numerical solvers and in physically realistic predictions of flood wave propagation and river discharge (Feng et al., 2023). PIML has also been applied in atmospheric sciences to simulate convection-resolving models, improving the temporal stability and physical understanding of learned patterns (Beucler et al., 2023; Mukkavilli et al., 2023). Although progress has been made, challenges persist, particularly in terms of training stability, scaling to high-dimensional problems, and compatibility with existing Earth system models. These restrictions highlight the need for a systematic analysis of existing practices and the determination of future research directions.

The current developments in PIML represent a paradigm shift in computational modeling, focusing on the interplay between domain knowledge and data-driven inference (Tkachov & Murashko, 2025; Yuan et al., 2025). PIML and PINN frameworks are capable of the previously mentioned computational intractability because of their ability to combine the interpretability of physics-based formulations with the flexibility of neural networks (Farias et al., 2025; Huynh, 2023). In weather prediction, for example, short-term rainfall prediction, parameter calibration in land-atmosphere interaction models, and bias correction in NWP output, PINNs have been investigated. They have been used in hydrology to study groundwater flow, sediment transport, infiltration, and flood forecasting (Habib et al., 2025; Zeinali et al., 2025). These applications demonstrate the flexibility of PINNs; however, they also highlight critical gaps in research, including the lack of field-scale validation experiments, quantification of uncertainty, and challenges in modeling interacting surface and subsurface processes.

Recent reviews in related fields can help overcome these challenges. Cai et al. (2021) surveyed the applications of PINNs to fluid mechanics, with a focus on inverse problems in which the boundary conditions or material properties are unknown, a

situation typical of atmospheric and hydrological systems (Cai et al., 2021). Similarly, Yu et al. (2025) followed the history of AI-based remote sensing inversion development, starting with physics-based models and moving to foundation models, and found that several issues remain, including interpretability, domain generalization, and uncertainty quantification (Yu et al., 2025). The findings can be integrated into the hydrometeorological setting to guide the construction of PINNs capable of processing high-dimensional spatiotemporal data, mixed observations and models, and complicated boundary interactions. In addition, the recent rise in the popularity of digital twins and hybrid physics-AI systems in Earth system surveillance (Chew et al., 2025) warrants further consideration. PINNs are poised to play a central role in enabling continuous, physically consistent data assimilation and forecast refinement (Chew et al., 2025).

Another critical motivation for this review is the lack of a unified taxonomy for evaluating PINN architectures in environmental applications. While structured frameworks have been developed for other fields—for example, the four-tier taxonomy proposed for aerospace engineering, weather, and hydrology—they lack comparable classification schemes (Tkachov & Murashko, 2025).

An existing gap in this area hinders systematic comparative analysis across studies and the identification of optimal practices for model design, training, and validation. Moreover, the literature on PINNs used alongside operational forecasting systems or ensemble modeling frameworks is also sparse. However, it goes without saying that physics-informed surrogates have the potential to reduce computational expenses in ensemble-based data assimilation and uncertainty propagation. As such, current synthesis methods and a consistent research agenda are needed to transform PINNs from experimental prototypes into operational hydrometeorological systems.

The objectives of this review are threefold: (1) to provide an overview of the current state of research on the application of PINNs in weather and hydrological models, to provide a summary of the techniques, structures, and performance outcomes; (2) It seeks to establish the theoretical and computational challenges that limit the scalability and reliability of PINNs in modeling nonlinear and multiscale processes in the environment. (3) To outline future directions and provide practical recommendations on how PINNs can be incorporated into current numerical models, remote sensing systems, and data assimilation frameworks to expand predictive abilities in time and space.

The review is organized around four questions: (1) What are the most commonly used PINN architectures and variants in weather and hydrological modeling? (2) What are the advantages and disadvantages of using PINN architectures and variants in weather and hydrological modeling? (3) What are the requirements for using PINN architectures and variants in weather and hydrological modeling? (4) What are the limitations of PINN architectures and variants in weather and hydrological modeling?

2. Method

2.1. Systematic literature review framework

The SLR was conducted using the Preferred Reporting Items for Systematic Reviews and Meta-Analyses framework (Haddaway et al., 2022). The framework was not evaluated but rather used to make the literature search transparent and reproducible. Scopus, Web of Science Core Collection, ScienceDirect, SpringerLink, IEEE Xplore, Taylor & Francis Online, MDPI, Google Scholar, and relevant publisher databases for hydrology, atmospheric science, and scientific machine learning were all searched. The last search was limited to English-language, peer-reviewed papers, conference papers, and high-relevance preprints from 2010 to 2025.

The full database search strings used were:

Scopus/Web of Science: TITLE-ABS-KEY(("physics-informed neural network*" OR PINN* OR "physics-informed machine learning" OR "scientific machine learning") AND (hydrolog* OR meteorolog* OR weather OR climate OR rainfall OR precipitation OR flood* OR groundwater OR "soil moisture" OR "shallow water" OR "Saint-Venant" OR Richards OR Navier-Stokes)) AND PUBYEAR > 2009 AND PUBYEAR < 2026 AND LANGUAGE(English); Google Scholar/publisher search:

("physics-informed neural network" OR PINN OR "physics-informed machine learning" OR "scientific machine learning") AND (weather OR meteorology OR hydrology OR flood OR rainfall OR groundwater OR "soil moisture" OR "shallow water" OR "Saint-Venant" OR Richards OR Navier-Stokes) AND (modeling OR forecasting OR prediction OR inverse OR surrogate). Truncation terms were used for hydrology*, flood*, model, forecast, and PINN*, which were supported by the database.

Filters were applied uniformly when they could be done at the database interface: English language, publication years (2010–2025), article/conference/review document types, and subject areas related to environmental science, water resources, geosciences, computer science, engineering, applied mathematics, and atmospheric science. Google Scholar was used as an ancillary source to gather preprints and recently accepted papers, with the first 200 relevance-ranked papers screened for reproducibility.

If a duplicate record was found, this was removed first during the screening process. One main reason that was given for the title and abstract exclusions was that it was not a PINN/PIML study, was not related to weather, climate, hydrology, or water-resources modeling, was not a modeling or methodological contribution, or was not available in English. The primary reason for assigning full-text exclusions was that there was not enough methodological detail, no explicit governing equation or physics constraint, no hydrometeorological application, no validation or performance metric, duplicate/preliminary version of an included study, or inaccessible full text. These categories were employed to ensure that the PRISMA flow diagram and the text matched, and to ensure the exclusion logic was auditable.

The study quality was evaluated based on four clear criteria: clarity of the description of the architecture, completeness of the reporting of the data sources, explicit specification of the governing equations and boundary/initial conditions, and completeness of the validation metrics or benchmark comparisons. Qualitative synthesis was used to retain the studies that were directly relevant to PINN/PIML hydrometeorological modeling and met two or more of the quality criteria. Studies that satisfied two or three of the four criteria were included in the category of comparatively well-validated studies; studies that satisfied two or three were included in the emerging evidence category; and studies that satisfied less than two were included in the category for contextual purposes only and were not used to support performance claims.

Table 1: Study-quality appraisal criteria for inclusion and synthesis weighting

Criterion	High-quality reporting	Moderate reporting	Use in synthesis
Architecture description	The PINN variant, inputs/outputs, network depth/width, loss terms, optimizer, and training strategy were specified.	The architecture is named, but the training or loss details are incomplete.	Determines whether architectural claims are treated as reproducible evidence.
Data sources	Observation, reanalysis, simulation, and experimental datasets are named with spatial/temporal coverage and preprocessing.	The dataset type is stated, but the coverage or preprocessing is unclear.	Determines whether the results are transferable to field-scale hydrometeorology.
Governing equations	PDE physical parameters, source terms, boundary conditions, and initial conditions are explicitly defined.	The equation family is named, but the symbols or constraints are partly undefined.	Determines whether the work is a physics-informed model or a generic neural network.
Validation metrics	Quantitative metrics and baselines are reported, such as RMSE, MAE, NSE, mass-balance error, conservation residual, or solver/runtime comparison.	Qualitative comparison or limited metric reporting only.	Determines whether efficiency, accuracy, and computational savings claims can be cited.

2.2. Data extraction and coding

All studies included were systematically extracted according to a predefined coding schema that provided bibliographic information (author(s), year, journal/conference/preprints, country of origin), modeling domain (atmospheric, hydrological, or coupled systems), and PINN architecture (feedforward, convolutional, recurrent, or hybrid with or without residual connections or physics-guided initialization).

2.3. Integration of PRISMA findings

This review, based on the PRISMA guidelines, presents a comprehensive qualitative synthesis of the methodological and application-specific issues of PINNs. This approach enabled (1) the classification of the most relevant publications in the PINN hydrometeorology sphere. (2) New trends, gaps in research, and under-explored areas of application mapping, such as coupled surface and subsurface hydrological modeling. (3) Methodological heterogeneity of studies, including the heterogeneity of PINN architectures, loss functional formulations, and evaluation metrics. (4) Determination of potential clusters that will be employed in the future in interdisciplinary collaboration and integration with working forecasting systems.

2.4. Limitations of the methodology

Although the PRISMA framework is methodologically rigorous, the review is limited to English-language publications, potentially missing studies in other languages. Moreover, bibliometric analysis relies on properly standardized metadata and keywords, which can introduce slight bias. Nevertheless, the PRISM method provides a sound and replicable methodology for assessing the application of PINNs to complex environmental systems.

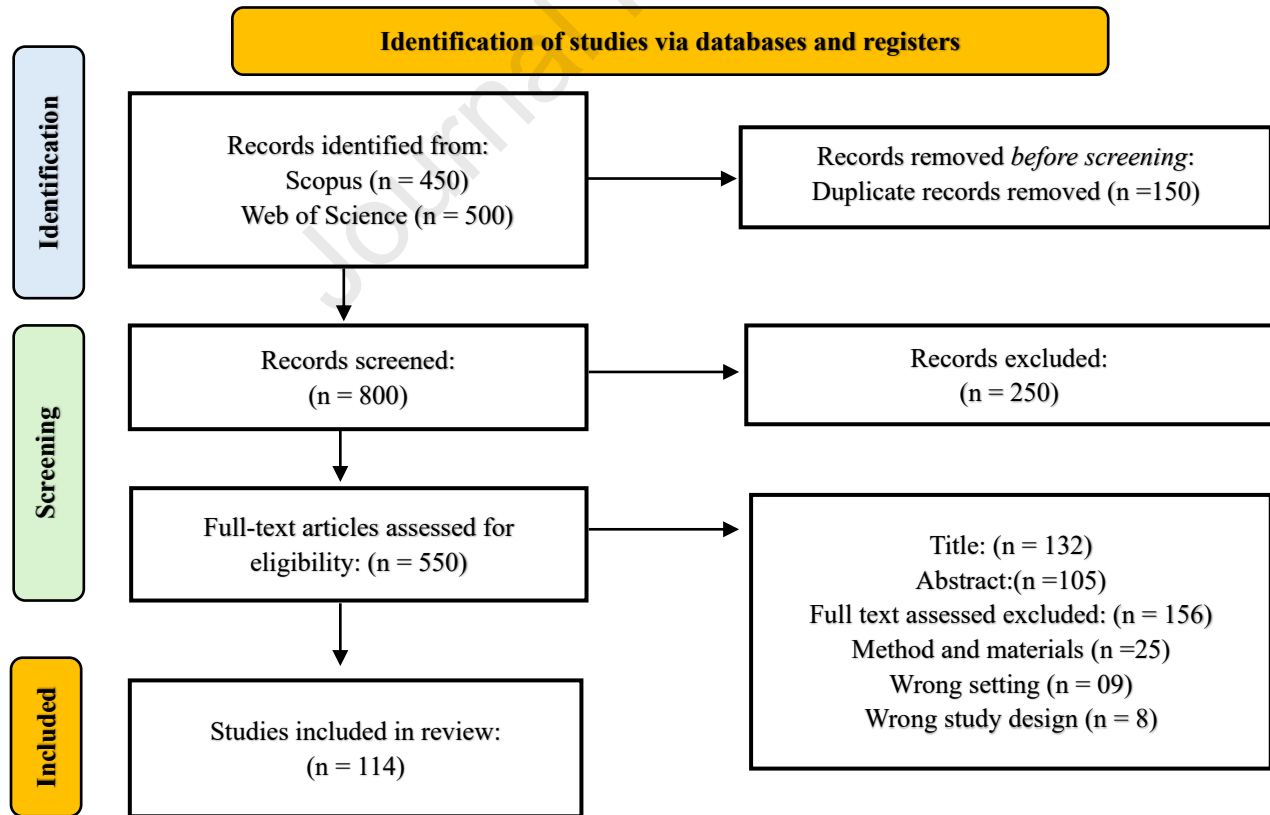


Fig. 1 Methodology for SLR for publication screening

3. Overview of PINN and its variants

This section provides a technical answer to the research question: What are the dominant modeling architectures used for PINNs and related methods in weather and hydrological sciences?

3.1. Overview of PINNs

PINNs are a set of DL models that directly incorporate physical laws (typically expressed as PDEs) into the neural network training objective (Huang & Wang, 2022; Raissi et al., 2019). The architecture of PINNs and the training process are illustrated in Fig. 2. (a) The conceptual framework combines neural network outputs with physical knowledge by incorporating PDEs and relevant physical conditions. (b) The network is optimized during training by minimizing a composite loss that combines data-constrained and physics-based residuals, thereby improving the model's consistency with governing hydrological and atmospheric processes. It improves the model's generalization to weather and hydrological predictions (Bandai & Ghezzehei, 2022; Feng et al., 2023).

Rather than training a purely data-driven mapping, a PINN constrains the hypothesis space so that the solution approximated is consistent with conservation laws, constitutive relationships, and other physical priors (Tartakovsky et al., 2020; Wang & Yang, 2024). PINNs successfully combine mechanistic modeling with statistical learning, enabling data-efficient and physically interpretable scientific ML.

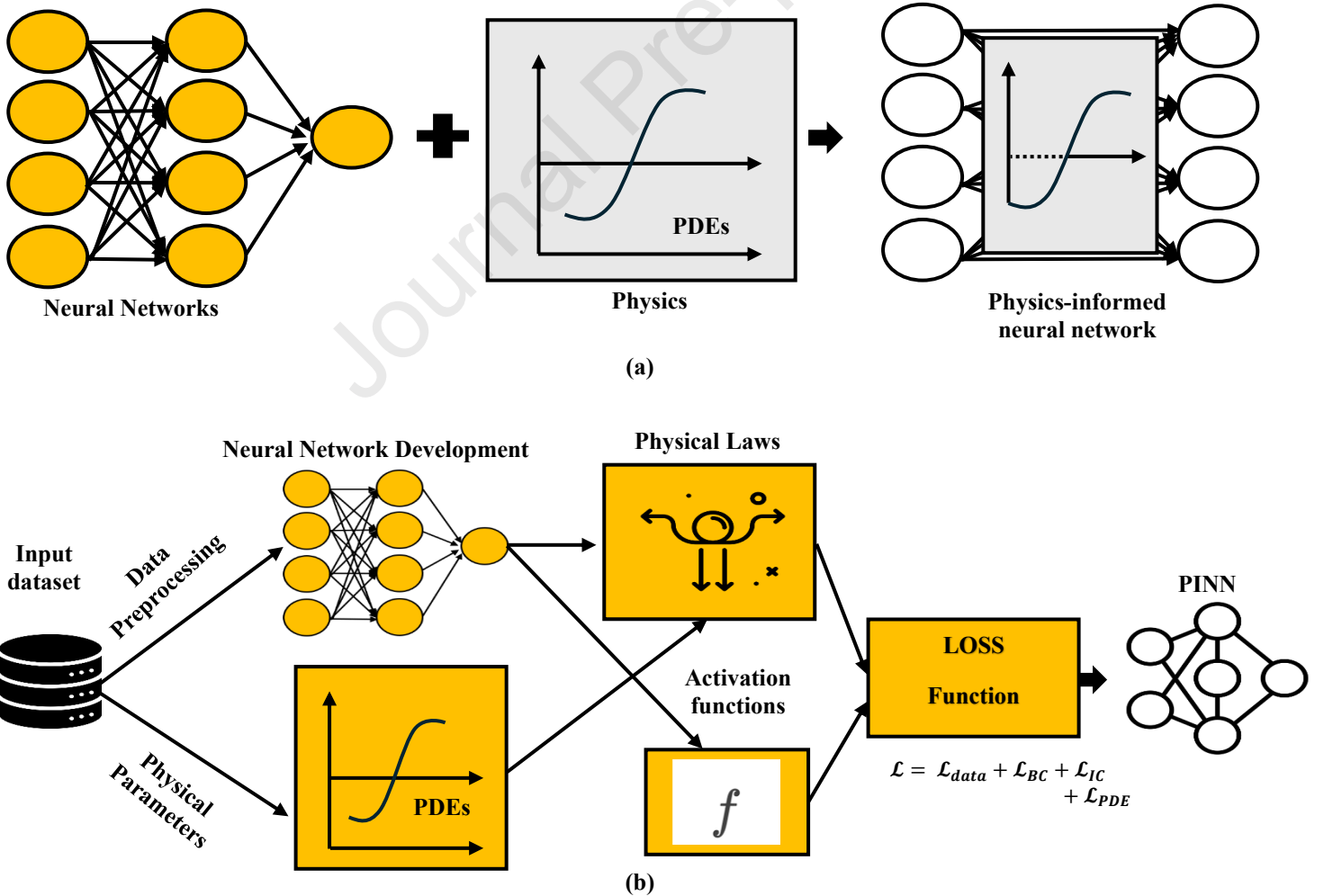


Fig. 2 (a) Conceptual framework of a PINN and **(b)** training and development of PINNs. The neural network learns from data and physical laws by embedding governing PDEs and boundary conditions into the training loss function.

Mathematically,

Let a general residual form of a PDE be a physical system expressed as $f[u(x, t); \lambda] - s(x, t) = 0$ for a spatial-temporal domain $\Omega \times [0, T]$. In this, u is the state variable that is approximated by the neural network, x and t are spatial and temporal coordinates, f is the differential operator, λ are physical parameters, and s is a source or forcing term.

$$\mathcal{N}[u(x, t), \lambda] - f(x, t), \quad x \in \Omega \subset \mathbb{R}^d, t \in [0, T] \quad (1)$$

subject to boundary and initial conditions

$$u(x, t) = g(x, t), \quad x \in \partial\Omega$$

$$u(x, 0) = h(x), \quad x \in \Omega$$

where $\mathcal{N}[\cdot]$ is a differential operator parameterized by physical coefficients λ , $f(x, t)$ denotes a source term, and $u(x, t)$ is the target field (e.g., velocity, temperature, or hydraulic head).

A standard PINN approximates $u(x, t)$ by a deep neural network $\hat{u}(x, t; \theta)$, with parameters $\theta = \{W_i, b_i\}_{i=1}^L$. Physics are enforced by penalizing the residual of the governing PDE, computed through automatic differentiation of the network output with respect to its inputs:

$$r(x, t; \theta) = \mathcal{N}[\hat{u}(x, t; \theta); \lambda] - f(x, t). \quad (2)$$

The composite loss function $\mathcal{L}(\theta)$ combines data fidelity and physics regularization terms:

$$\mathcal{L}(\theta) = \mathcal{L}_{data} + \mathcal{L}_{BC} + \mathcal{L}_{IC} + \mathcal{L}_{PDE} \quad (3)$$

Where,

$$\mathcal{L}_{data} = \frac{1}{N_d} \sum_{i=1}^{N_d} \|\hat{u}(x_i, t_i; \theta) - u_i^{obs}\|^2,$$

$$\mathcal{L}_{BC} = \frac{1}{N_b} \sum_{i=1}^{N_b} \|\hat{u}(x_i, t_i; \theta) - g(x_i, t_i)\|^2,$$

$$\mathcal{L}_{IC} = \frac{1}{N_0} \sum_{i=1}^{N_0} \|\hat{u}(x_i, 0; \theta) - h(x_i)\|^2,$$

Moreover, the physics residual term is expressed as follows:

$$\mathcal{L}_{PDE} = \frac{1}{N_f} \sum_{i=1}^{N_f} \|r(x_i, t_i; \theta)\|^2.$$

The optimization problem is thus formulated as

$$\theta^* = \arg \min_{\theta} \mathcal{L}(\theta). \quad (4)$$

It is typically solved with stochastic gradient descent or adaptive gradient algorithms such as Adam. The inclusion of $\mathcal{L}_{PDE}$ constrains the learned function to be consistent with physical laws at collocation points $\{x_i, t_i\}_{i=1}^{N_f}$, effectively transforming the learning task into a physics-regularized regression problem.

In forward modeling, the physics parameters λ are known, and u is predicted by the network. Inverse problems, with parameters λ considered learnable variables, enable the network to simultaneously learn the system parameters and the state field by minimizing the physics-informed loss (shown in Fig. 5(a and b)).

3.2. Different variants of PINNs in weather and hydrological applications

3.2.1. Vanilla PINNs

Even though flexible, traditional PINNs (also referred to as vanilla PINNs) have convergence issues in stiff, multi-scale, and highly non-linear problems (Cuomo et al., 2022; DUTTA & DAS, 2025). The competition between the gradient scales of the data-fitting loss and the PDE-residual loss is likely to cause an imbalance in the minima, leading to suboptimal minima (Medeiros Davi, 2022). Moreover, when global collocation is employed, high-dimensional domains or long-time integrations can become computationally expensive (Wang & Yang, 2024). Consequently, numerous PINN variants have been proposed to improve numerical stability, scalability, and accuracy.

In atmospheric sciences and hydrology, a vanilla PINN is implemented as a fully connected multilayer perceptron (MLP) that takes spatial-temporal coordinates (x, t) (and possibly forcing variables) as inputs and outputs the target state variable $\hat{u}(x, t; \theta)$ (e.g., hydraulic head, velocity, pressure, moisture). The governing PDE residual

$$r(x, t; \theta) = N[\hat{u}(x, t; \theta); \lambda] - f(x, t). \quad (5)$$

Where $N[\cdot]$ represents the differential operator and λ denotes physical parameters. This residual is enforced via automatic differentiation, and the total loss is:

$$\mathcal{L}(\theta) = \mathcal{L}_{data} + \mathcal{L}_{BC} + \mathcal{L}_{IC} + \mathcal{L}_{PDE} \quad (6)$$

With

$$\mathcal{L}_{PDE} = \frac{1}{N_f} \sum_{i=1}^{N_f} \|r(x_i, t_i; \theta)\|^2. \quad (7)$$

The data, boundary condition (BC), and initial condition (IC) terms are in standard formulations. In forward problems, λ is fixed, and only θ is optimized; in inverse problems, λ is also a learnable parameter, and thus both the field and underlying parameters are learned at the same time (Raissi et al., 2019). Vanilla PINNs have been applied to the shallow-water equations on the sphere (Bihlo & Popovych, 2022) and to learn fine-scale wind and pressure fields using sparse observations by including a Navier-Stokes regularization term in $\mathcal{L}_{PDE}$ (Soto et al., 2024). Flood-wave propagation and channel-flow modeling have seen

the use of vanilla PINNs, especially in early research that has combined data and PDE solvers (Nazari, 2025; Nazari et al., 2022).

Nevertheless, these field-specific vanilla PINNs frequently face stiffness issues, multi-scale behavior, unbalanced loss gradients, and long-time integration errors (DUTTA & DAS, 2025). The magnitudes of the gradient terms for the data loss and PDE residual terms often conflict, slowing convergence or leading to a poor minimum. Moreover, as the domain dimension or time horizon increases, global collocation sampling becomes computationally expensive and inefficient in capturing local fine-scale structures (Cuomo et al., 2022). For example, Vanilla PINN architecture is shown in Fig. 3, where the MLP produces $\hat{u}$. It is differentiated automatically to calculate residuals, and the various loss functions are then combined (Leng & Thiyagalingam, 2022).

Vanilla PINNs serve as a reference architecture that is smooth, low-dimensional, and moderately nonlinear in these regimes. However, they usually require improvement, such as through adaptive weighting, domain decomposition, or hybrid loss constructions, to be effectively applied in hydrological and weather modeling.

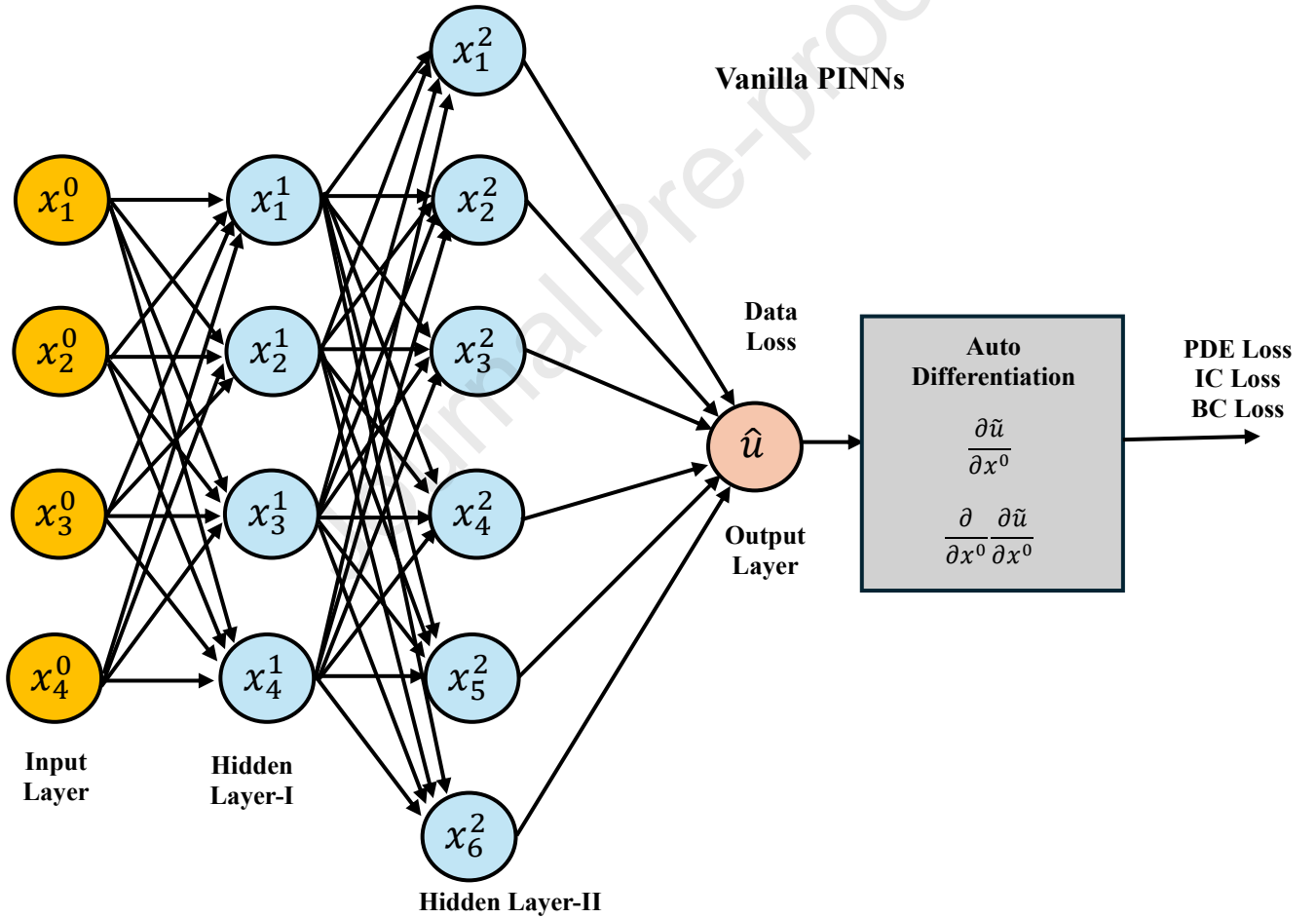


Fig. 3 Architecture of Vanilla PINNs (Leng & Thiyagalingam, 2022).

3.2.2. Variational PINN

The loss of the variational PINN (vPINN) is re-defined on the weak (integral) form of the governing equation rather than the strong form of the differential equation (Kharazmi et al., 2021). In the vPINN, a weak (integral) residual of the pointwise

(strong-form) PDE is used in the loss, similar to a Petrov–Galerkin formulation of the variational problem (Kharazmi et al., 2019).

Instead of enforcing

$$r(x, t; \theta) = N[\hat{u}(x, t; \theta); \lambda] - f(x, t) \quad (8)$$

to be (approximately) zero at collocation points, VPINN is defined for a test function space $\{\phi_j(x, t)\}$ the weak residuals

$$R_j(\theta) = \iint_{\Omega \times [0, T]} (N[\hat{u}] - f) \phi_j d\Omega dt - (\text{boundary/IC surface terms}) \quad (9)$$

Furthermore, this minimizes the variational loss.

$$L_{var}(\theta) = \sum_j |R_j(\theta)|^2. \quad (10)$$

Using integration by parts, the higher-order derivatives of $\hat{u}$ are moved onto the test functions ϕ_j , which makes the network less differentiable and removes numerical stiffness (i.e., the order of differentiation within the network) (Kharazmi et al., 2021).

vPINNs have demonstrated strong capabilities for hydrological and atmospheric PDEs, including advection-diffusion, the Richards equation, groundwater flow, and shallow water equations (Jia et al., 2025). In contrast to conventional PINNs, which apply PDE residuals pointwise, vPINNs have a weak-form formulation, enabling sharp and discontinuous features (e.g., in hydraulic head or flux) to be better captured through residual enforcement using an integral (Wang et al., 2025b).

The traditional vPINN is a weakly defined formulation that is also less sensitive to collocation noise, as a test-function-based residual is averaged over space and time, thereby removing local errors (Kharazmi et al., 2021; Liu & Wu, 2023). The hp-vPINN extension subdivides the computational domain into subregions of local polynomial test spaces to improve convergence and reduce parameter correlation, which can be applied to heterogeneous hydrological domains, such as river networks (Kharazmi et al., 2021; Kharazmi et al., 2019).

Meshfree vPINN designs are even more versatile than vPINNs, dynamically sampling test functions based on error estimates and concentrating computation where it is needed (Berrone & Pintore, 2024). Nevertheless, there are still problems, such as the intricate layout of the test function spaces, the high computational cost of numerical quadrature, and the inability to implement boundary and initial conditions using the weak formulations (Berrone et al., 2023).

Nevertheless, vPINNs are a strong and promising extension of regular PINNs to more intricate hydrological and atmospheric contexts, offering greater flexibility, stability, and accuracy than traditional models (Wang et al., 2023). The vPINN schematic architecture is depicted in Fig. 4, with the neural network outcome $\hat{u}_{NN}$ assessed and derivatives $d\hat{u}_{NN}$ and derivatives of test functions $d\phi$ computed to create the variational residual that determines convergence (True/False) by its weak-form loss (Kumar & Singh, 2025).

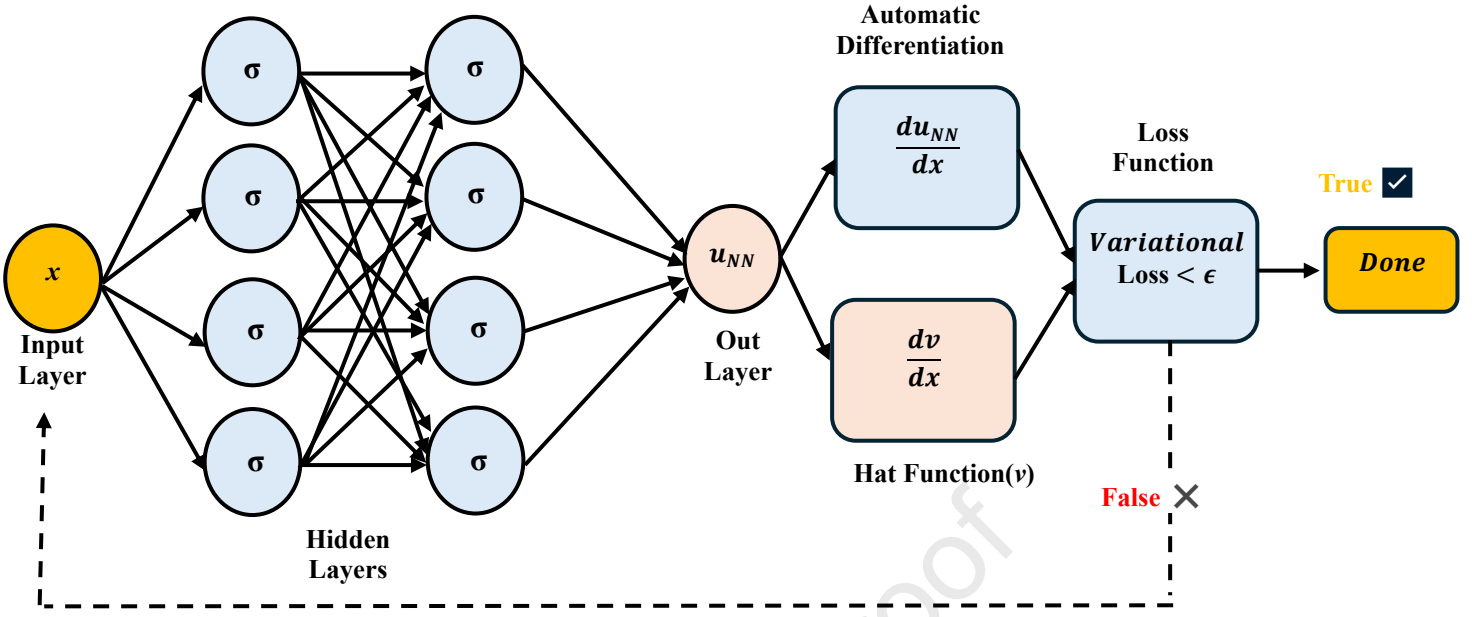


Fig. 4 Architecture of the variational PINN (Kumar & Singh, 2025).

3.2.3. Extended PINN

Extended PINNs (XPINNs) generalize the PINN framework by decomposing the whole spatio-temporal domain $\Omega \times [0, T]$ into multiple subdomains $\{\Omega_k\}_{k=1}^K$, each endowed with its own neural network $\hat{u}_k(x, t; \theta_k)$ (Hu et al., 2021; Jagtap & Karniadakis, 2020; Ren et al., 2024). This domain decomposition can be executed in space, time, or both—providing flexibility and potential parallelism beyond cPINNs (Shukla et al., 2021).

The loss in the original XPINN formulation (Jagtap & Karniadakis, 2020) consists of subdomain data losses, subdomain residual losses, and interface-continuity terms that are averaged over interface collocation points. Thus, the interface contribution needs to be in the form of an average mismatch at the shared interface rather than in the form of an unnormalized sum. The XPINNs were created as a general space-time domain-decomposition approach, which is parallelizable and can be used for complex geometries; subsequently, a number of studies were carried out to apply the approach to more specific interface problems with modified interface conditions and loss terms.

$$r_k = (x, t; \theta_k) = N[\hat{u}_k(x, t; \theta_k)] - f(x, t). \quad (11)$$

It is a local loss

$$\mathcal{L}_k = \mathcal{L}_{k,data} + \mathcal{L}_{k,BC} + \mathcal{L}_{k,IC} + \mathcal{L}_{k,PDE}. \quad (12)$$

The global loss becomes.

$$\mathcal{L}_{XPINN} = \sum_{k=1}^K \mathcal{L}_k - \mathcal{L}_{interface}. \quad (13)$$

$\mathcal{L}_{interface}$ enforces continuity across subdomain interfaces $\Gamma_{k,k+1}$:

$$\hat{u}_k = \hat{u}_{k+1}, \quad \nabla \hat{u}_k \cdot \mathbf{n}_k = \nabla \hat{u}_{k+1} \cdot \mathbf{n}_{k+1} \quad \text{on } \Gamma_{k,k+1}. \quad (14)$$

These interface constraints preserve the value continuity and flux (gradient) consistency within the subdomains.

Computationally, XPINNs have two key advantages: (1) parallel training of the K sub-networks, which is less time-consuming, particularly with large K , and (2) the complexity of adaptive models per subdomain-stiff gradient or heavily heterogeneous regions can be allocated deeper or coarse networks and more intensive collocation sampling, and smoother regions can be allocated light models (Jagtap & Karniadakis, 2020).

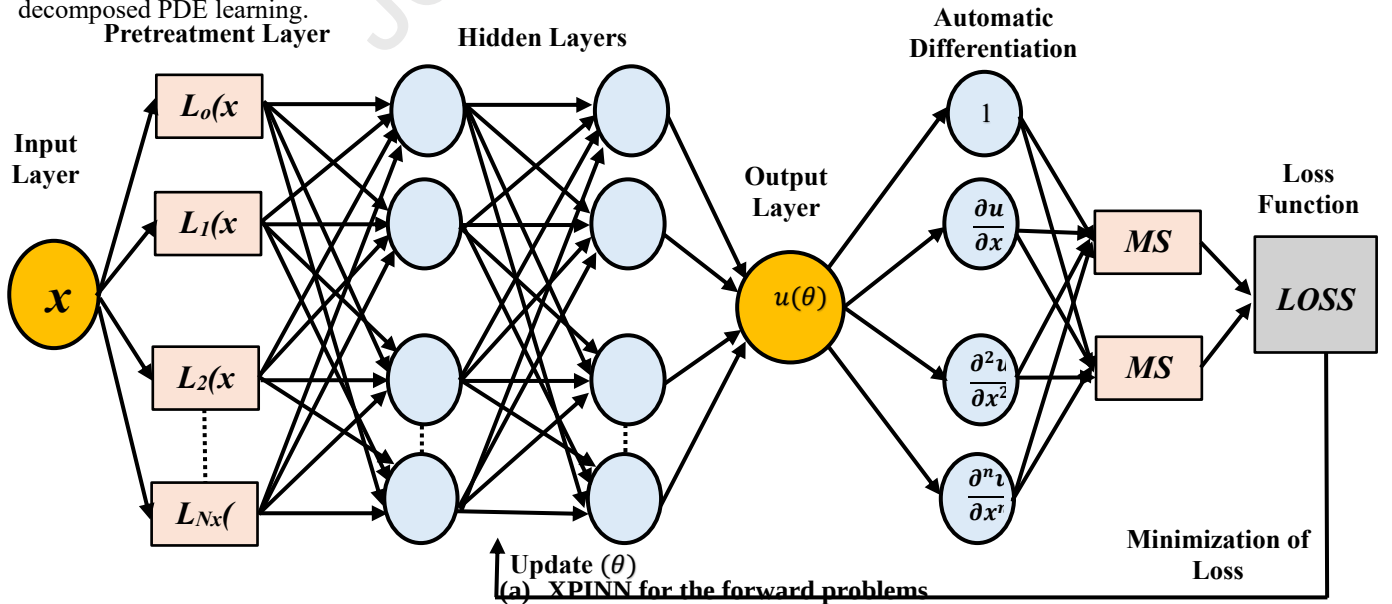
XPINNs are especially well-suited for modeling heterogeneities in hydrological and atmospheric models (e.g., layered strata and variable conductivity) and aligning physical discontinuities with subdomain boundaries (Hu et al., 2021; Jagtap & Karniadakis, 2020; Zhang et al., 2024). For example, in porous-flow problems with spatially varying permeability, the porous domain is divided into nearly homogeneous regions, each of which is solved by a sub-PINN, and the continuity of the pressure and flux across the interfaces is enforced (Alamu et al., 2025; Jagtap & Karniadakis, 2020).

Weather modeling can use a space-time decomposition, in which physical regimes (e.g., boundary layer vs. free troposphere) are more or less adaptive and can be applied (Alamu et al., 2025; Kashinath et al., 2021). Recent discoveries in shallow-water and atmospheric PDEs indicate that XPINNs can minimize spectral bias and training stiffness, thereby improving the accuracy and convergence (Bihlo & Popovych, 2022; Shukla et al., 2021).

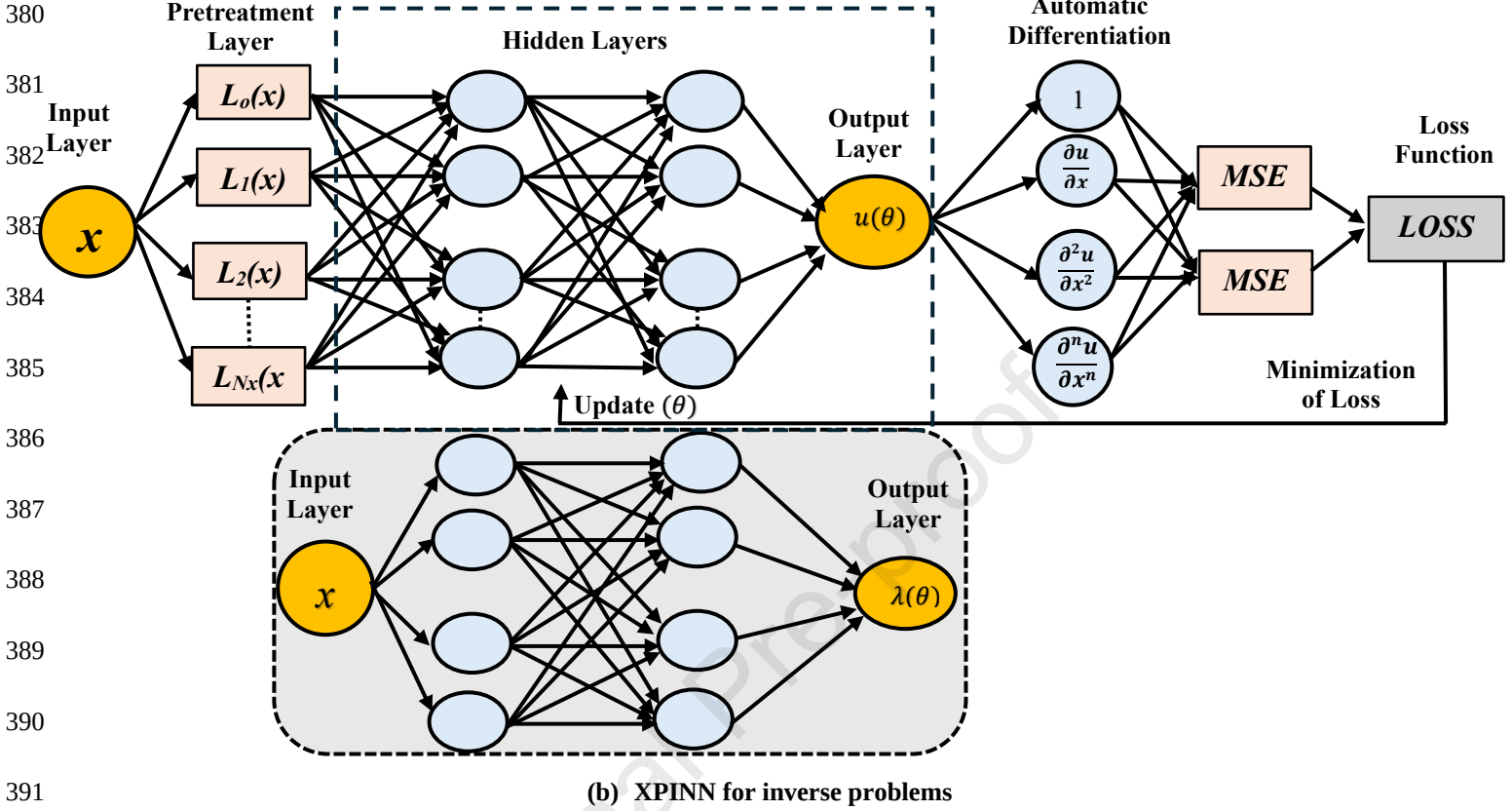
Nevertheless, XPINNs incur communication overhead due to interface coupling during training; domain partitioning (size, shape, and number of subdomains) is a nontrivial choice (Hu et al., 2021; Shukla et al., 2021). Hu et al. (2021) conducted an analytical study that explored the timing of XPINNs, which demonstrated better generalization in contrast to single-network PINNs using various decomposition strategies (Hu et al., 2021).

However, XPINNs have become a superior and scalable addition to the PINN family, offering higher parallelism, adaptivity, and resilience to complex hydrological and atmospheric systems. Figs. 5 (a) and (b) demonstrate a schematic XPINN architecture, which depicts the domain decomposition, local sub-PINNs, and continuity of the interface, which, in combination, represent the XPINN learning process (Zhang et al., 2024).

Fig. 5 is not a specific interface-problem method but a domain-decomposition workflow. The entire domain is divided into space-time subdomains; local neural networks are used to model each subdomain, and consistency is maintained by interface residuals and data/physics losses. To use it for interface-problem applications, extra interface conditions or modified losses must be applied, and the original XPINN formulation was introduced to enhance flexibility and parallelization in space-time decomposed PDE learning.



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Fig. 5 Architectural differences between XPINNs; (a) XPINN for the forward problem, and (b) XPINN for the inverse problem (Zhang et al., 2024).

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3.2.4. Conservative PINN

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The conservative PINN (cPINN) is also similar; however, a conservative representation of the governing equations imposes flux continuity at the edges of the subdomains (Hu et al., 2023; Wang et al., 2022a). The self-adaptive PINN (SA-PINN) proposes a dynamic weighting of the composite loss, meaning that the weight of each term depends on the magnitude of its gradient or the proportion of the loss's variance (Braga-Neto, 2021; McClenny & Braga-Neto, 2023). This modification addresses the imbalance between the physics and data terms, effectively serving as an equivalent to self-attention in learning and focusing on under-resolved regions (McClenny & Braga-Neto, 2023).

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Conservation laws (mass, momentum, and energy) have a natural divergence (flux) formulation in several hydrological and atmospheric PDEs (Naz et al., 2008). The continuity equation is a classical equation.

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$$\frac{\partial U}{\partial t} + \nabla \cdot F(U) = S, \quad (15)$$

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or in a more general variable-coefficient form

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$$\frac{\partial U}{\partial t} + \nabla \cdot (F_c(U) - D(U, \nabla U)) = S, \quad (16)$$

where F_c is the convective flux, D is the diffusive flux, and S is the source/sink. The cPINN architecture enforces flux continuity across subdomain interfaces, preserving the conservation property in a strongly enforced fashion (Jagtap & Karniadakis, 2020).

In cPINN, the computational domain Ω is partitioned into non-overlapping subdomains $\{\Omega_p\}_{p=1}^{N_{sd}}$. Within each subdomain, a local neural network $u^{(p)}(x, t; \theta^{(p)})$ is trained. The loss in each domain includes (i) data term, (ii) PDE residual, (iii) boundary/initial conditions, plus interface coupling terms that enforce:

Flux continuity:

$$(F_c - D)^{(p)} \cdot n_p = (F_c - D)^{(q)} \cdot n_q. \quad (17)$$

On the interface between subdomains p and q , in the strong form.

Solution averaging/matching: An interface consistency constraint is often imposed.

$$u_{interface}^{(p)} = \frac{1}{2} (\hat{u}^{(p)} - \hat{u}^{(q)}). \quad (18)$$

Alternatively, a penalty can be used to enforce the continuity of the state variable (Jagtap et al. 2020).

Hence, the composite loss in each subdomain becomes.

$$\mathcal{L}^{(p)} = \mathcal{L}_{data}^{(p)} + \mathcal{L}_{PDE}^{(p)} + \mathcal{L}_{PDE}^{(p)} + \mathcal{L}_{BC/IC}^{(p)} + \mathcal{L}_{interface}^{(p)}, \quad (19)$$

Moreover, global training is sought.

$$\{\theta^{(p)}\} = \underset{\{\theta^{(p)}\}}{\operatorname{argmin}} \sum_p L^{(p)}. \quad (20)$$

Fig. 6 shows a schematic of the cPINN architecture that decomposes the subdomains and local neural solvers and enforces conservation laws at the interface (Jagtap & Karniadakis, 2020).

Because the flux balance is strictly maintained, cPINNs can maintain mass, momentum, and energy balances, which are essential for hydrological fluxes and the atmospheric mass balance. Domain decomposition also addresses training issues such as stiffness and gradient conflicts and allows variable network depth and width per subregion (Bhasme et al., 2022; Jagtap & Karniadakis, 2020; Shukla et al., 2021).

cPINNs can enforce continuity of water flux across geological or topographic boundaries in hydrological applications. cPINNs allow mass or momentum conservation across grid patches or layers in vertical systems in the atmosphere (Nazari, 2025; Nazari et al., 2022; Secci et al., 2024). Nevertheless, cPINNs have not been widely used in hydrology and weather modeling, with most studies being exploratory. cPINNs are primarily strong in conservation structure, numerical drift reduction, and error propagation through domain partitions.

Fig. 6 emphasizes the conservative extension: local subnetworks solve subdomain equations, whereas interface penalties enforce flux continuity and conservation across subdomain boundaries. This distinction is important because cPINNs focus more directly on the conservation law structure, whereas XPINNs provide a broader decomposition framework.

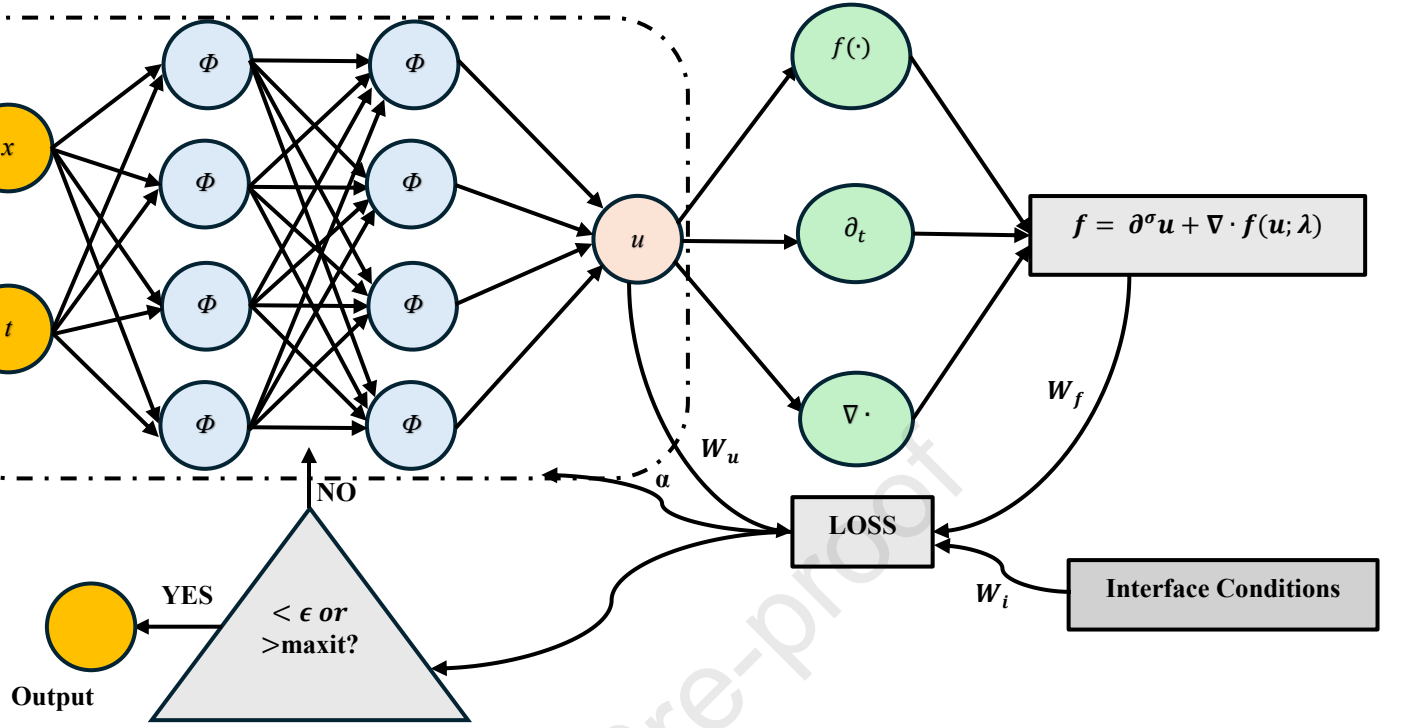


Fig. 6 Conservative PINNs architecture (Jagtap & Karniadakis, 2020).

3.2.5. Fractional PINNs

Fractional PINNs (fPINNs) generalize standard PINNs to settings governed by fractional (nonlocal or memory) differential operators, such as time-fractional derivatives or spatial fractional Laplacians (Pang et al., 2019; Wang et al., 2025a). Such formulations are attractive in hydrology and atmospheric modeling because many transport, diffusion, and retention processes exhibit anomalous (e.g., sub- or super-diffusive) behavior and extended memory (e.g., soil moisture retention, pollutant transport, and turbulence) that integer-order PDEs may inadequately capture.

A typical fractional PDE system might read:

$$D_t^\alpha u(x, t) + \kappa(-\Delta)^{\beta/2} u(x, t) = f(x, t), \quad x \in \Omega, t \in [0, T], \quad (21)$$

with initial and boundary conditions. Here D_t^α is a Caputo (or Riemann–Liouville) time-fractional derivative of order $0 < \alpha \leq 1$ and $(-\Delta)^{\beta/2}$ is the fractional Laplacian (e.g., Riesz form) of order β . The corresponding fPINNs define a residual:

$$r_f(x, t; \theta) = D_t^\alpha \hat{u}(x, t; \theta) + \kappa(-\Delta)^{\beta/2} \hat{u}(x, t; \theta) - f(x, t). \quad (22)$$

Because automatic differentiation does not directly generalize to fractional operators (fractional derivatives do not follow the standard chain rule), fPINNs often hybridize autodiff for integer-order portions and numerical discretization (e.g., Grünwald–Letnikov, spectral, or finite-difference approximations) for fractional parts (Sivalingam et al., 2025). Thus, the total loss might combine

$$\mathcal{L} = \mathcal{L}_{data} + \mathcal{L}_{BC} + \mathcal{L}_{IC} + \mathcal{L}_{fracPDE}. \quad (23)$$

Where,

$$\mathcal{L}_{fracPDE} = \frac{1}{N_f} \sum_{i=1}^{N_f} \|r_f(x_i, t_i; \theta)\|^2.$$

One advanced variant is adaptive fractional PINNs, which dynamically adjust collocation points or fractional orders during training to improve convergence in complex domains (Ma et al., 2025). Another is the time-difference PINN (TD-PINN), which discretizes the time-fractional operator (e.g., BDF2 + L1 formula) while retaining a neural network representation in space, making it suitable for fractional water-wave and shallow-water models (Liu & Wu, 2023; Liu et al., 2023).

In hydrology, fPINNs have been applied (or proposed) for unsaturated flow processes with power-law diffusion and retention kernels, leveraging fractional orders to represent memory in soil moisture dynamics (Wang et al., 2025a; Wang et al., 2025b). In atmospheric contexts, fractional forms may better capture nonlocal mixing and anomalous dispersion of aerosols or pollutants, although explicit implementations remain emergent (Suzuki et al., 2023). In practice, training leverages collocation sets, boundary/initial constraints, and hybrid numerical schemes to enforce fractional operator fidelity (Wang et al., 2022b). The promise lies in richer physical expressivity, better long-term memory capture, and parameter discovery (e.g., inferring fractional orders and diffusion coefficients) in inverse problems, although challenges in training stability, computational cost, and scalability remain active research fronts (Ma et al., 2025; Pang et al., 2019).

The dynamics of the model are fractional or memory dependent, as shown in Fig. 7, and are represented by fractional derivative operators in the residual term. This is important in hydrological applications for anomalous transport, long-memory soil-water movement, and nonlocal subsurface processes; however, it is not yet widely used at the field scale because of limited parameter identifiability and validation data.

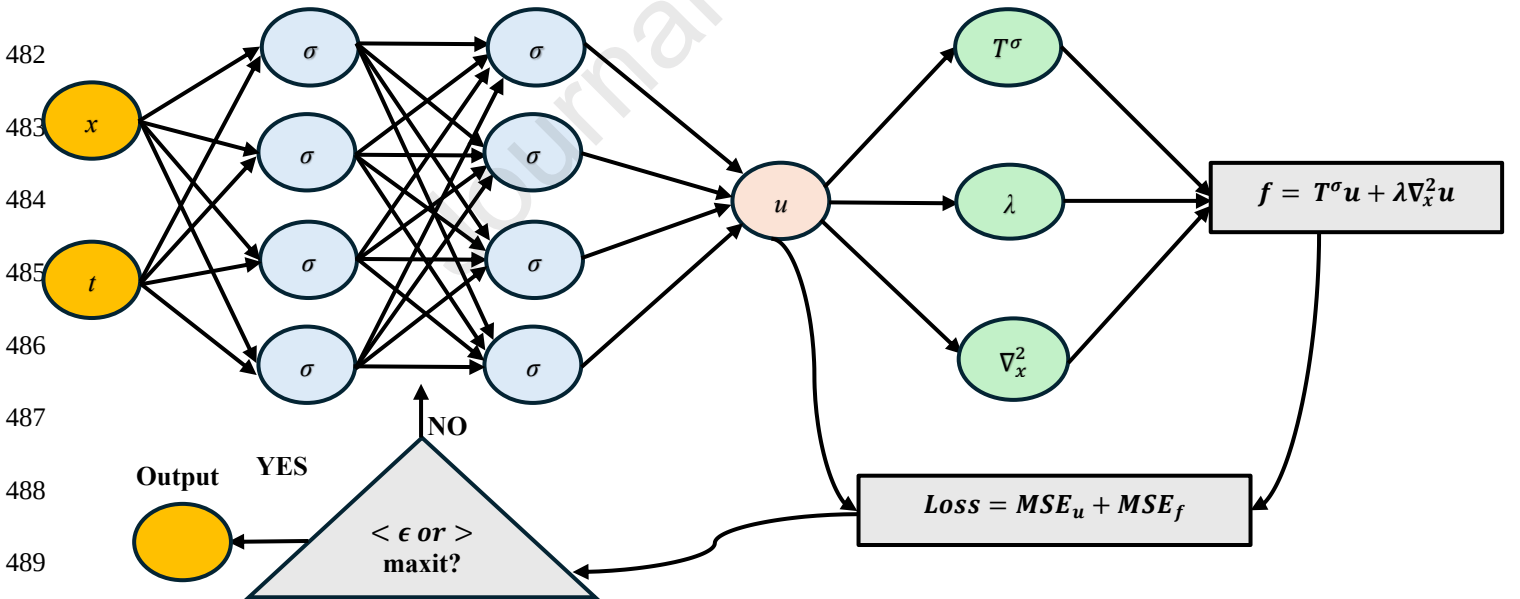


Fig. 7 Architecture of conformable time-fractional PINN (Wang et al., 2022b).

3.2.6. Stochastic PINNs

Stochastic variability occurs in many geophysical and hydrological processes, such as solute transport in groundwater, soil moisture diffusion, and atmospheric dispersion, which are driven by non-uniform subsurface structures, uncertain boundary conditions, or random forcing. Stochastic PINNs (sPINNs) extend classical deterministic PINN models by incorporating

randomness through a network architecture that involves stochastic processes and quantifies uncertainty (Chen et al., 2021; Habib et al., 2025).

The control stochastic partial differential equation (SPDE) may be formulated as follows:

$$\mathcal{L}[u(x, t; \xi)] = f(x, t; \xi), \quad (x, t) \in \Omega \times [0, T], \quad (24)$$

where ξ denotes a vector of random variables representing uncertain inputs (e.g., diffusivity, velocity). The stochastic field $u(x, t; \xi)$ is typically expanded via Karhunen–Loève (KL) or polynomial chaos (PC) representations:

$$u(x, t; \xi) = \sum_{k=0}^K \hat{u}_k(x, t) \Phi_k(\xi) \quad (25)$$

where $\Phi_k(\xi)$ are orthogonal PC basis functions, and $\hat{u}_k(x, t)$ are deterministic coefficients learned by neural networks.

The total residual is minimized by enforcing the SPDE and boundary/initial constraints across the stochastic domain as follows:

$$\mathcal{L}_{SPINN} = \mathbb{E}_{\xi}[\|\mathcal{L}[u_{\theta}(x, t; \xi)]\|^2] + \lambda_{BC/IC} \mathcal{L}_{BC/IC} \quad (26)$$

with Monte Carlo estimation of the expectation or quadrature integration. Practically, multiple neural networks, one neural network per stochastic mode or field component, are used.

Chen et al. (2021) demonstrated an approach to stochastic advection–diffusion–reaction systems, where the field of mean diffusivity and coefficients of the stochastic concentration response are modeled with independent NNs (Chen et al., 2021). ID variations are arbitrary polynomial chaos PINNs (aPC-PINNs), which improve the model of non-Gaussian randomness, and multi-fidelity sPINNs, which integrate low- and high-resolution observations with composite networks (Meng & Karniadakis, 2020; Meng et al., 2020). Subsequent models utilize Gaussian process surrogates to accelerate uncertainty propagation and employ Bayesian PINNs for probabilistic inference (Yuan et al., 2025).

Despite this promise, sPINNs are computationally expensive and suffer from high sampling costs and instability in variance, particularly with high-dimensional random inputs. Adaptive collocation techniques, spectral dropout regularization, and physics-directed priors are being studied to minimize overfitting and improve convergence (Zeinali et al., 2025).

Stochastic PINN learning under uncertainty is shown in Fig. 8, but the concept of stochastic PINNs should be understood broadly. Bayesian neural networks, Bayesian PINNs, physics-informed priors, stochastic differential equation constraints, ensemble PINNs, and meta-learning approaches are all included in the literature. These approaches provide more explicit quantification of epistemic and aleatory uncertainty than deterministic PINNs; however, they are typically more costly to compute.

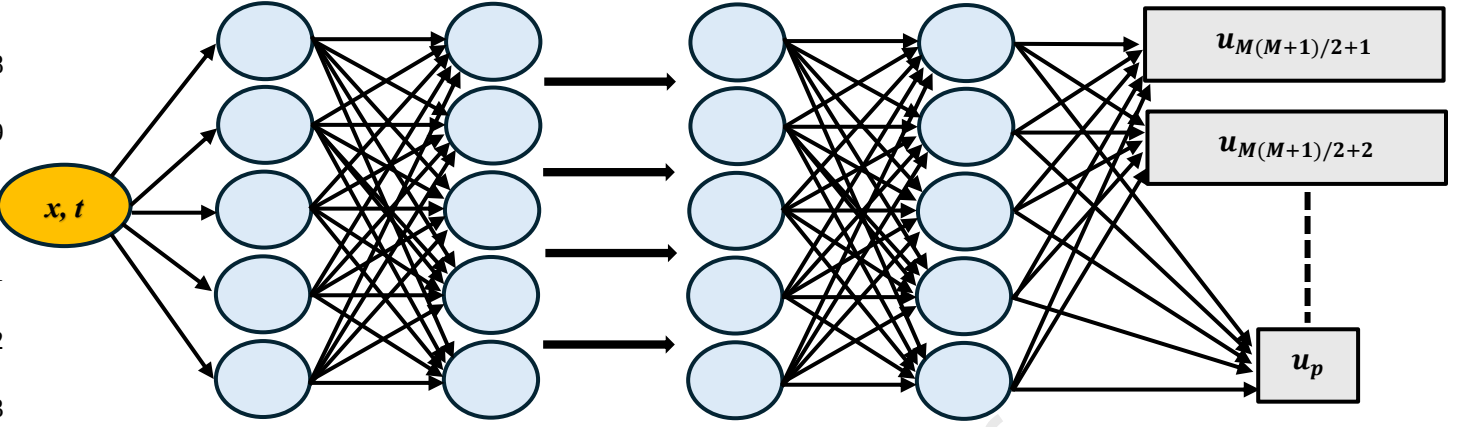


Fig. 8 Architecture of stochastic PINNs (Chen et al., 2021).

3.2.7. Mesh-free PINNs

Conventional numerical methods based on PDEs, including finite element or finite volume methods, require meshing, which is challenging in complex or dynamic geophysical regions (e.g., irregular watersheds or dynamic floodplains) (Nazari, 2025; Nazari et al., 2022). Mesh-free PINNs (MF-PINNs) address this issue by directly learning solutions at randomly sampled collocation points, which makes them appealing for hydrology and weather modeling of irregular geometries (Raissi et al., 2019; Zeinali et al., 2025).

The MF-PINNs can be modeled as

$$\mathcal{N}[u_\theta(x, t)] = f(x, t), \quad (x, t) \in \Omega \times [0, T], \quad (27)$$

and the neural network $u_\theta(x, t)$ approximates the unknown field without any mesh discretization. The loss function ($\mathcal{L}$) combines data mismatch, physical residuals, and boundary/initial penalties:

$$\mathcal{L}(\theta) = \mathcal{L}_{data} + \lambda_1 \mathcal{L}_{BC/IC} + \lambda_2 \frac{1}{N_f} \sum_{i=1}^{N_f} \|u_\theta(x_i, t_i) - f(x_i, t_i)\|^2. \quad (28)$$

The mesh-free technique consists of randomly sampling training points (x_i, t_i) through the domain. Automatic differentiation uses spatial and temporal derivatives and does not rely on explicit numerical grids.

The method is beneficial in free-surface flow and coupled groundwater–surface water problems, where the domain changes with time. Zeinali et al. (2025) demonstrated that MF-PINNs founded on the shallow water equations (SWE) can generate dynamic flood wave propagation in sparse measurements, which are superior to the conventional solvers in terms of generalization and speed (Zeinali et al., 2025).

More advanced mesh-free methods include domain decomposition PINNs (DD-PINNs), which scale domains by decomposing them into smaller sub-networks (Jagtap & Karniadakis, 2020). MF-PINNs are also valuable for inverse problems (e.g., inference of fields of hydraulic conductivity) because they can flexibly represent boundaries and are easily coupled with measured information. However, issues persist with sampling density selection, training rigidity, and edge error localization.

MF-PINNs offer an alternative to hydrological and atmospheric PDEs, providing both geometric and mesh independence from the grid. They offer high levels of flexibility for empirical geoscientific problems with difficult or evolving boundaries,

providing an intermediate solution between solvers based on traditional methods and physics-based learning from data, and rely on automatic differentiation and adaptive collocation (Berrone & Pintore, 2024).

3.3. Case studies in weather and hydrology

Table 2: Comparative synthesis of major PINN variants in weather and hydrological modeling

PINN variant	Typical domains	Governing equations	Data sources	Strengths	Limitations / evidence maturity
Vanilla PINN	Groundwater, shallow-water flow, heat/moisture transport	Darcy, Richards, Saint-Venant, Navier-Stokes, advection-diffusion	Synthetic PDE data, sparse sensors, reanalysis, laboratory/field data	Simple formulation; useful for forward and inverse problems with sparse labels	Sensitive to loss weighting and stiffness; evidence strongest in controlled and low-dimensional settings
XPINN / domain decomposition	Heterogeneous aquifers, river channels, multi-scale domains	Subdomain PDE residuals with interface continuity	Simulation outputs, sparse observations, collocation points	Improves parallelization, handles irregular domains and discontinuities	Interface losses require careful averaging and tuning; field validation remains limited
cPINN	Conservative transport, flood waves, coupled flow systems	Conservation-law residuals and flux-continuity constraints	Numerical benchmarks, hydraulic observations	Better mass/flux conservation across subdomains	More complex interface enforcement; limited operational demonstrations
vPINN / hp-VPINN	Weak-form PDE learning, noisy or sparse data	Variational/weak forms of governing PDEs	Collocation points, finite-element-like test functions, observations	Can improve stability and reduce derivative-order demands	Implementation is more demanding; performance depends on test-space choices
Fractional / nonlocal PINN	Anomalous transport, memory-dependent subsurface processes	Fractional diffusion, nonlocal Laplacian, fractional water-wave models	Synthetic and limited experimental data	Represents memory and nonlocality not captured by classical PDEs	Parameter identifiability and field-scale validation are still early-stage
Stochastic / Bayesian PINN	Uncertain forecasts, sparse measurements, inverse problems	SDE/PDE residuals with probabilistic priors or posterior inference	Sparse sensors, ensembles, synthetic stochastic systems	Provides uncertainty quantification and physics-informed priors	Higher computational burden; hydrometeorological applications remain proof-of-concept

The emergence of PINNs has recently become diverse across a range of hydrological and atmospheric systems, enabling data-efficient and physically accurate modeling of nonlinear processes (DUTTA & DAS, 2025; Guo et al., 2023; Kharazmi et al., 2021; Wang & Yang, 2024). Standard and enhanced PINNs have been applied to river hydraulics, subsurface flow, infiltration, and atmospheric dynamics, and each uses governing equations specific to the domain.

This section reviews PINNs and their different variants used in weather and hydrological modeling (Table 1). Feng et al. (2023) applied a standard PINN with a Fourier embedding to scale down the magnitude of river flow on the Saint-Venant equations and to achieve a more accurate sub-grid water-depth prediction (Feng et al., 2023). Similarly, PINNs have been used to model a shallow-water system and the HEC-RAS system by Qi et al. (2024) and Zoch et al. (2025), respectively, and were demonstrated to be capable of making high-speed predictions and being physically consistent when modeling an open-channel system (Qi et al., 2024; Zoch et al., 2025). Domain-decomposition PINNs have also been successful at handling discontinuities and heterogeneities in hydraulic conductivity in unsaturated and groundwater flow (Bandai & Ghezzehei, 2022), and data-

574 fusion and multi-stage versions were more stable and precise for inverting non-linear infiltration problems (Li et al., 2025;
575 Sakar et al., 2024).

576 The strengths of hybrid and inverse PINNs have been applied in subsurface parameter estimation. Groundwater PINNs were
577 used by Zhang et al. (2022, 2023) and Guo et al. (2023) to estimate the parameters at the boundary, transmissivity, and to delete
578 non-uniqueness in the scenario of sparse data (Guo et al., 2023; Zhang et al., 2023; Zhang et al., 2022). Stochastic extensions
579 of Karhunen-Loève expansions were used to reconstruct spatially varying conductivity fields, such as KLE-PINN (Ji et al.,
580 2024). Soto et al. (2024) demonstrated that PINNs with Navier-Stokes equations could be relevant to the representation of
581 mesoscale weather fields (Soto et al., 2024), but Brecht and Bihlo (2024) used simplified PINNs with simple numerical weather
582 prediction systems of the last century (Brecht & Bihlo, 2024). Notably, Pang et al. (2019) introduced the concept of fPINNs to
583 model anomalous diffusion and memory in hydrological and atmospheric transport, and extended PINNs to encompass nonlocal
584 and memory-length processes (Pang et al., 2019). These studies demonstrate the scope of PINNs as an integrative paradigm
585 that combines physical principles and ML to predict the behavior of complex geophysical systems.

586 Table 3: Overview of PINN and variants applications in hydrology and weather modeling

Study	PINN Variant / Approach	Governing Equation(s)	Domain / Application	Key Features / Contributions
(Feng et al., 2023)	Standard PINN + Fourier embeddings	1D Saint-Venant (mass + momentum)	River flows downscaling / open channel	PDE-constrained downscaling; Fourier features for tidal BCs; improved subgrid water-depth prediction vs. interpolation.
(Bandai & Ghezzehei, 2022)	Domain-decomposition PINN	Richards' equation	Unsaturated/layered soils	Domain splitting for discontinuous K; adaptive activation; inverse estimation of BCs; and parameters from sparse data.
(Zhang et al., 2022)	GW-PINN	Darcy + continuity	Groundwater/aquifers	Hard/soft constraints; solves forward and inverse head/flux; no labeled data; identifies seepage/boundary parameters.
(Li et al., 2025)	Data-fusion/dual-driven PINN + Hydrus-1D	Richards' infiltration equation	Unsaturated infiltration	The PINN is combined with Hydrus data to stabilize the training, manage nonlinear diffusion, and operate in a data-scarce regime.
(Sakar et al., 2024)	Multi-stage PINN	Saturation + petrophysical relations	2D heterogeneous infiltration	Sequential training (saturation → permeability → refinement); geoelectrical tomography integration; robust inverse mapping.
(Shadab et al., 2023)	Steady unconfined-flow PINN	Dupuit–Boussinesq / Di Nucci models	Phreatic surface / free-surface flow	Estimates hydraulic conductivity and free-surface profile; assessed under measurement noise; robustness analysis.
(Zoch et al., 2025)	Surrogate/operational PINN	1D/2D Saint-Venant (PDE in HEC-RAS)	River stage forecasting	PINN surrogate of HEC-RAS; enforces PDE; high-speed inference; local deviations in some reaches.
(Zhang et al., 2023)	Parameter-ID PINN	Darcy + continuity	Aquifer seepage/flow	Hard-constraint PINN for velocity + conductivity ID; forward + inverse unified framework; geotechnical validation
(Guo et al., 2023)	HT-PINN	Groundwater flow PDEs	Hydraulic tomography	Physics-regularized inversion of transmissivity mitigates non-uniqueness and is efficient in high-dimensional settings.

(Ji et al., 2024)	KLE-PINN	Groundwater flow + K–L expansion	Groundwater inverse	Combines PINN + KLE for stochastic K-field reconstruction and limited-data inversion demonstration.
(Depina et al., 2022)	Standard/inverse PINN	Richards' equation	Unsaturated zone	Inverse estimation of unsaturated parameters: improved stability versus purely data-driven models.
(Secci et al., 2024)	Transient unconfined-flow PINN	Free-surface groundwater PDEs	Phreatic surface dynamics	PINN handles moving free-surface BCs; transient simulation validated against analytical/numerical models.
(Hu et al., 2025)	Physics-regularized PINN	Richards' equation	Near-surface soil moisture	Embedded soil physics constraints, sparse sensor assimilation, and rapid predictive inference.
(Qi et al., 2024)	Standard PINN	2D shallow-water equations	Free-surface / open-channel flow	Solves non-linear SWEs for variable Froude regimes; robustness and numerical stability analysis.
(Soto et al., 2024)	Atmospheric PINN	Navier–Stokes (momentum + mass)	Mesoscale meteorological fields	Reconstructs high-resolution wind and pressure from sparse data; ensures physical consistency.
(Brecht & Bihlo, 2024)	Simplified/historical PINN	Early NWP dynamical equations	Historical weather (ENIAC)	Reproduces early NWP forecasts with modern PINN surrogates and demonstrates physics-guided learning.
(Pang et al., 2019)	fPINNs (fractional PINN)	Fractional advection–diffusion	Hydrological/ atmospheric transport	Extends PINNs to Caputo/Riesz derivatives; models anomalous diffusion and memory effects
(Zhu et al., 2025)	PINN surrogate with forcings	Shallow-water / storm-surge PDEs	Coastal/ surge prediction	Physics-informed surrogate for storm surge under limited tide–wind forcing; operational forecasting potential

3.4. Difference between PINNs and CNNs

It is not necessary to view PINNs and CNNs as competing model classes. CNNs are designed for gridded data, while PINNs are designed by training and regularization principles, where physical equations, constraints, or priors are added to the loss function, architecture, or parameterization. Physics-informed CNNs, convolutional surrogates with PDE residual losses, and vision models with physical priors are thus valid hybrids. If dense gridded observations are the dominant feature of the problem, or if reanalysis products are used, CNNs are often a better choice, whereas if physical constraints, sparse labels, inverse estimation, extrapolation beyond the training data, or conservation laws are more relevant, PINNs are more appealing.

PINNs and CNNs are distinct concepts within a DL framework for environmental modeling (Jia et al., 2025). Although they can be classified as part of the broader subcategory of neural architecture, they are distinguished by significant differences in design, training objectives, and mathematical concepts (Cerniauskas et al., 2024; Minadakis, 2024). CNNs are more data-driven learning models, which are based on statistical learning on a large dataset (Waqas et al., 2024; Waqas et al., 2023), unlike PINNs that explicitly include physical laws in the form of PDEs in the learning process (Cai et al., 2021; DUTTA & DAS, 2025; Jagtap & Karniadakis, 2020; Kharazmi et al., 2021; Pang et al., 2020). These differences are essential for weather, climate, and hydrological modeling.

CNNs were initially developed to operate on grid-based spatial data by exploiting local spatial correlations using convolutional filters and weight sharing (Yu & Ji, 2019). This design also enforces translation equivariance, allowing for the efficient detection of features (edges, textures, or spatial gradients) in an image or field (LeCun et al., 2015; Yu & Ji, 2019).

CNNs have proven helpful in geophysical modeling for forecasting precipitation, creating subsurface satellite-based maps, and anticipating hydrological conditions, with support from exceedingly large datasets (Liu et al., 2020; Shahriar et al., 2025; Yu et al., 2022). The network is trained to learn a mapping $f_0: X \rightarrow Y$. However, CNNs do not assume the physics of the system; therefore, they only generalize to the space of the training data (Tsourounis et al., 2022).

In comparison, a PINN integrates the guiding PDEs into the training problem, transforming the regression problem into a physics-regularized optimization problem (Huang & Wang, 2022; Raissi et al., 2019). PINN networks are not only trained on data but also minimize a composite loss that includes data misfit, boundary, and initial condition errors, as well as residuals from the governing PDE (Farea et al., 2024; Khalid et al., 2024). The spatial and temporal derivatives can also be correctly calculated with respect to the inputs using automatic differentiation, ensuring that the solution obtained is consistent with the conservation laws and physical constraints (Wang & Yang, 2024). PINNs can effectively extrapolate to thus far unmonitored spatiotemporal areas, which is one of the principal advantages of hydrological and meteorological prognostication; the information is often incomplete or unevenly distributed (Farea et al., 2024; Khalid et al., 2024).

There operate at various architectural, structural, and computational levels. CNNs are discrete, grid-based, and hierarchical, and use local receptive fields and convolutional kernels to extract spatial patterns (Malik et al., 2023). In contrast, PINNs are usually continuous, mesh-free, and founded on fully connected MLPs, which are expected to model continuous functions with constant domains (Zhao et al., 2024a). The implications of these architectural differences extend well beyond representational strength, training behavior and computational efficiency.

Furthermore, Fig. 9 shows a CNN architecture for differentiating from PINNs (Fig. 2). Overall, a CNN consists of convolutional and max pooling layers, followed by fully connected layers, where the feature maps gradually decrease in size and flatten. The feature maps are then classified using dense layers with an activation function.

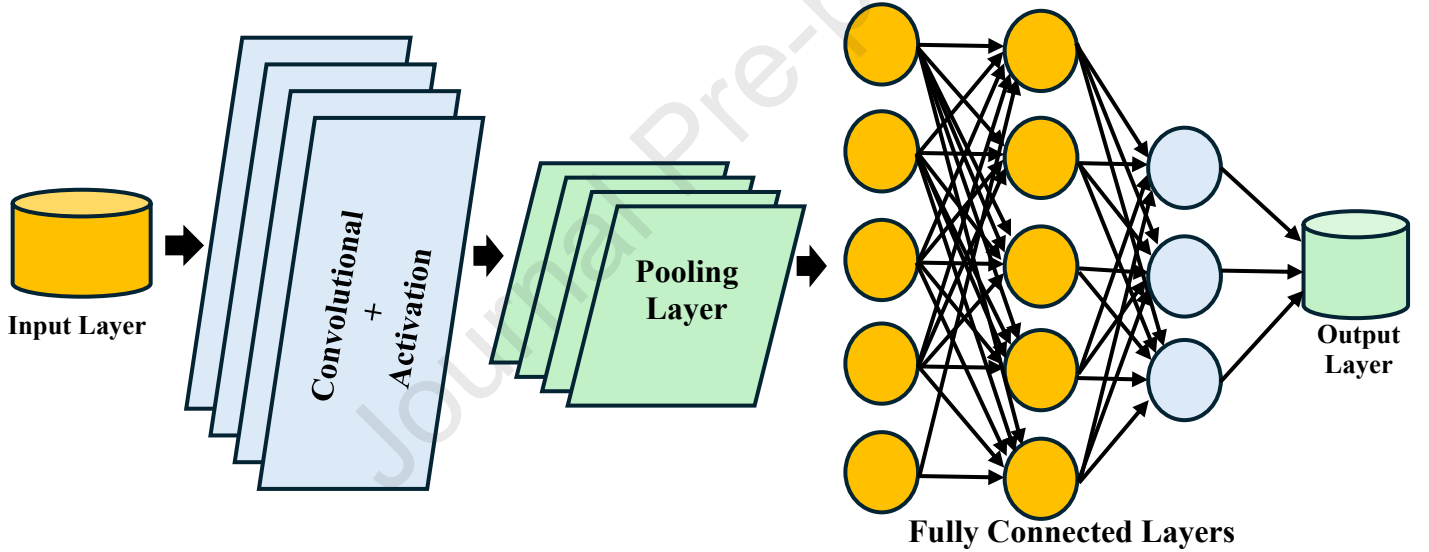


Fig. 9 General architecture of CNNs

A comparative approach to architectural and functional aspects is summarized in Table 4. CNNs are highly adapted to structured, data-rich worlds to achieve spatial correlations at computational efficiency, but at the cost of deplorable physical interpretability (Malik et al., 2023). However, PINNs are often used as continuous functional approximators when the PDEs are provided and sufficient discretization is applied (Miyagawa & Yokota, 2024; Zhao et al., 2024a). Generalization and physical fidelity can be more effectively tackled by PINNs, which are more costly to compute than canonical or parametric models in the case of weather and hydrological models, where the governing equations are sufficiently well understood and observations are sparse (Donnelly et al., 2024; Nazari, 2025; Nazari et al., 2022). Physics-informed CNNs and graph-based networks are also designed as hybrid architectures that combine space-efficient CNNs with physics awareness (Feng et al., 2025; Zhou et al., 2024).

Table 4: Architectural and functional comparison between CNNs, PINNs, and physics-informed hybrid CNN/PINN models

Aspect	CNNs	PINNs
Primary Objective	It learns a data-driven mapping ($f_\theta: X \rightarrow Y$) that minimizes empirical loss over labeled datasets.	Solve or approximate solutions to governing physical equations by minimizing both data and PDE residual losses in a unified framework.
Architectural Paradigm	Hierarchical grid-based architecture composed of convolutional, pooling, and fully connected layers. Employs local receptive fields and weight sharing for spatial feature extraction.	Typically, a fully connected MLP architecture that maps continuous coordinates (x, t) to field variables $u(x, t)$. Each neuron operates globally, with no spatial locality assumption.
Mathematical Representation	$y = \sigma(W * x + b)$ denotes discrete convolution and operates on tensor data.	$\hat{u}(x, t; \theta) = f_\theta(x, t)$, where f_θ is a smooth function approximator; $\frac{\partial u}{\partial x_i}$ and $\frac{\partial^2 u}{\partial t^2}$ are computed analytically using automatic differentiation.
Architectural Bias	Incorporates inductive biases such as translation equivariance, local connectivity, and spatial invariance, which are ideal for structured (grid) data, such as satellite images or radar maps.	The architecture assumes functional smoothness and continuous differentiability to ensure compatibility with the PDE residual computation and does not assume translation invariance.
Input/Output Representation	The inputs and outputs are typically fixed-size tensors (e.g., 2D or 3D images). The architecture of the proposed model is inherently discrete.	Inputs are continuous spatial-temporal coordinates, and outputs are continuously valued physical fields (e.g., pressure, velocity, temperature, or hydraulic head).
Core Computational Mechanism	Uses discrete convolution and pooling operations to reduce dimensionality and capture hierarchical features; efficient on GPUs using tensor operations.	Relies on automatic differentiation to compute the PDE residual. $r(x, t; \theta) = \mathcal{N}[\hat{u}(x, t), \theta] - f(x, t)$; optimization enforces physics constraints via collocation points.
Loss Function Structure	Primarily data-driven: $\mathcal{L}_{\text{data}} = \frac{1}{N} \sum_i \ y_i - \hat{y}_i\ ^2.$	Composite loss: $\mathcal{L} = \mathcal{L}_{\text{data}} + \mathcal{L}_{\text{BC}} + \mathcal{L}_{\text{IC}} + \mathcal{L}_{\text{PDE}}$, combining data, boundary, initial, and physics residual terms.
Spatial/Temporal Modeling Capability	Captures local spatial dependencies effectively but lacks explicit temporal or physics constraints unless extended (e.g., ConvLSTM, 3D-CNN).	Intrinsically, models spatiotemporal evolution governed by PDEs; temporal and spatial dependencies are enforced through physical operators.
Domain Geometry Handling	Performs best on regular Cartesian grids and requires remeshing or interpolation for irregular domains.	It is naturally mesh-free and can sample collocation points arbitrarily in complex or unstructured domains.
Training Mechanism	Trained using stochastic gradient descent on large datasets; gradients computed through backpropagation over convolutional layers.	Trained via gradient-based optimization (e.g., Adam, L-BFGS), minimizing PDE residuals, and relying heavily on automatic differentiation for PDE evaluation.
Parameter Learning (Inverse Problems)	Learns phenomenological mappings and is not inherently suited for estimating physical parameters.	Can jointly infer system parameters λ and field solutions $u(x, t)$ by treating λ as trainable variables in the physics-informed loss.
Generalization / Extrapolation	Extrapolation beyond data boundaries is often unstable, limited to the data distribution seen during training.	Physics constraints enhance the extrapolation and robustness, particularly in data-sparse or unseen regions.

Computational Efficiency	Highly parallelizable: convolution operations optimized on GPUs. Efficient inference after training.	Computationally expensive during training owing to PDE residual evaluations; scalability remains a research challenge.
Activation Functions	ReLU, LeakyReLU, and GELU are preferred for sparse activations and fast convergence.	Smooth and differentiable activations (e.g., tanh, sin, and Swish) are preferred to ensure derivative continuity for the PDE terms.
Output Interpretation	It produces discrete feature maps or class probabilities. Interpretability is mainly statistical or pattern-based in nature.	Produces physically interpretable continuous fields, which can be interpreted through governing equations and conservation laws.
Hybridization Potential	It can be augmented with physics-inspired constraints (e.g., loss regularization or PDE residual penalties).	Can integrate convolutional blocks (PICNNs, PhyGeoNet) for spatial feature extraction within physics-informed frameworks.

4. Discussion on the study's research questions

The rapid evolution of PINNs has transformed the way environmental scientists approach nonlinear and multi-scale problems in weather and hydrological modeling. This section describes the findings of this review in relation to the research questions that drive the study, taking into consideration the methodological implications of the new developments and cross-domain applications.

4.1. Dominant modeling architectures used for PINNs and their variants

This study highlighted current PINN architectures, including vanilla PINNs, vPINNs, XPINNs, cPINNs, and sPINNs, all optimized for particular categories of physical and computational problems (Hu et al., 2021; Jagtap & Karniadakis, 2020; Kharazmi et al., 2021; Nazari, 2025; Raissi et al., 2019; Shukla et al., 2021). Vanilla PINNs are basic yet versatile and user-friendly PINNs that utilize an MLP to reduce a composite loss, incorporating data, boundary, and physical residual terms (Bandai & Ghezzehei, 2022; Feng et al., 2023). However, it tends to cause convergence instability in stiff or multi-scale problems when enforcing PDE residuals.

To address these issues, vPINNs reformulate the loss in the weak (integral) form, thereby simplifying numerical conditioning and the handling of discontinuities in hydraulic head or flux within heterogeneous hydrological systems (Jia et al., 2025; Kharazmi et al., 2021; Kharazmi et al., 2019). Similarly, XPINNs develop this concept using domain decomposition, parallel training of subdomains, and interface continuity (Jagtap & Karniadakis, 2020). Specifically, this decomposition enables the modeling of extremely heterogeneous systems, such as stratified aquifers or heterogeneous layers with different stability in the atmosphere, in a manner that conserves the conservation laws at the interface boundaries, which is an essential consideration in the surface-subsurface hydrological interaction of the atmosphere and the surface (Alhubail et al., 2022; Hu et al., 2021; Secci et al., 2024; Zhang et al., 2024).

The uncertainty in sPINNs injects random variables into the network, which can be tracked within a physics-based framework (Chen et al., 2021; Habib et al., 2025). With these architectures, the impact of uncertain inputs, such as variable rainfall, soil heterogeneity, and boundary fluxes, is measured to provide physically consistent estimates of uncertainty. Fractional and mesh-free PINNs go further, with the variants of the models being capable of modeling memory and nonlocal transport processes (Ma et al., 2025; Pang et al., 2019), and mesh-free models not necessitating a grid to be generated in irregular regions, such as floodplains and changing watersheds (Zeinali et al., 2025).

Overall, these structural developments suggest a hybridization tendency: spatial adaptivity, stochastic representation, and domain decomposition (Ji et al., 2024). Hybrid PINNs are also taking over as surrogate solvers for numerical solvers, achieving the same level of accuracy at a much lower computational cost (Donnelly et al., 2024; Nazari, 2025; Nazari et al., 2022).

The range of these architectures reflects the flexibility of the PINN paradigm in managing the equations that govern environmental systems, spanning Navier-Stokes-based atmospheric models to the Richards equation in unsaturated hydrology.

4.2. Reconciling data-driven learning and physical constraints

The reviewed literature suggests that PINNs can be successfully used in the data-driven learning/physical constraints procedure to develop composite loss functions that reconcile the requirements of empirical and theoretical consistency (Farea et al., 2024). This two-sidedness enables PINNs to act as physics-regularized regressors, ensuring robust predictions that can be interpreted even when the observational data are sparse or noisy (Habib et al., 2025).

The challenge of data scarcity and its heterogeneity is a classical issue in hydrology, whereas governing equations, such as the Darcy law, Saint-Venant system, or Richards equation, can be used to regularize learning with PINNs (Bandai & Ghezzehei, 2022; Feng et al., 2023). PINNs have been used across the atmospheric sciences, where nonlinearity and chaos are expected, and have been applied to the computation of convection-resolving and downscale coarse-resolution forecasts without mass or energy conservation (Beucler et al., 2023). The indication of such applications suggests that physics-directed priority can be combined with sparse sensor measurements or model outputs using PINNs, resulting in superior generalization and reduced uncertainty.

However, the trade-off between the database and physics-based term of the loss is a technical problem. Overweighting of physics residuals can impede learning in noisy domains, whereas underweighting can yield physically erratic answers (Braga-Neto, 2021; McClenny & Braga-Neto, 2023). Adaptive weighting strategies, self-attention PINNs, and vPINNs have been proposed as solutions to this imbalance by dynamically adjusting the role of each loss term during training (Braga-Neto, 2021; Kharazmi et al., 2021). PINNs can be used with data assimilation models, such as ensemble Kalman filters and four-dimensional variational models, to combine real-time measurements with physics-constrained learning to produce improved forecasts (Hu et al., 2025; Kashinath et al., 2021).

4.3. Limitations and prospects of implementation of PINNs/variants

In terms of operational forecasting, PINNs are best suited as hybrid rather than stand-alone elements of numerical weather prediction or hydrological models. The practical applications involve (i) using PINNs as an expensive subgrid process surrogate; (ii) coupling PINNs with ensemble Kalman filters, 3D-Var, or 4D-Var to assimilate sparse station, radar, satellite, and reanalysis data; (iii) embedding PINN modules within the hydrological digital twins for near real-time state estimation; and (iv) using PINNs to reconstruct missing fields and poorly observed parameters, such as hydraulic conductivity, roughness, infiltration capacity, and boundary fluxes.

The deployment restrictions are still very high. Automatic differentiation can be slower than traditional neural networks because it requires repeating the evaluation of PDE residuals at numerous collocation points. The availability of data is variable over basins and events, boundaries are frequently unknown, and operational systems must have strict latency, reliability, calibration, and uncertainty communication requirements. For this reason, PINNs are best considered decision-support and model-augmentation tools at this time, unless they have been compared to operational observations and local numerical solvers within the target lead time and spatial resolution.

While promising, PINNs have challenges associated with the technique as well as hydrometeorological deployment. The technique-level limitations comprise training instability for stiff and multi-scale PDEs, spectral bias, gradient imbalance between data and physics losses, sensitivity to collocation-point sampling, hard boundary conditions, and computational cost due to automatic differentiation. The hydrometeorological limitations are due to uncertainties in forcing and boundary conditions, noisy and sparse field observations, parameter equifinality, spatial variability, and lack of validation in high-dimensional forecasts at basin, regional, or continental scales.

PINNs are also susceptible to convergence instability methodologically in stiff and multi-scale systems, whose scales of the loss components are of different magnitudes (Wang & Yang, 2024). Systems that are ill-posed in inverse problems, where the parameters of the system being learned are determined along with the state variables, can have multiple minima that represent

physically distinct solutions (Depina et al., 2022). Another frontier actively pursued is uncertainty quantification, which is addressed through stochastic extensions (sPINNs) and Bayesian PINNs that capture probabilistic parameter distributions (Ji et al., 2024; Yuan et al., 2025).

There is a need to qualify performance claims. There are several potential advantages of PINNs, including a decreased amount of labelled data needed for training or reduced training time after training; however, computational advantages are problem-dependent and should be backed up by quantifiable metrics such as wall clock training time, inference time, memory usage, conservation residuals, RMSE/MAE/NSE, or a comparison with calibrated numerical solvers and purely data-driven baselines. Algorithmic gains rely heavily on the physical problem, data regime, and optimization approach, as highlighted by related recent work for robust PINNs and intelligent optimization of thermal/fluid systems (Abdulkarim et al., 2026; Demir et al., 2025; Nooruldeen et al., 2025), and the significance of operational data representation and climate-relevant decision contexts is highlighted in wider studies of climate and energy systems (Salman et al., 2021; Salman et al., 2025).

Although the potential of PINNs in weather and hydrological modeling is apparent, several technical, methodological, and operational limitations exist. The most important limitations are associated with computational cost, convergence instability, data sparsity, and scalability, whereas the opportunities are hybrid algorithmic integration, uncertainty quantification, and foundation-model integration. The limitations and emerging prospects are summarized in Table 3.

Table 5: Limitations and prospects of PINN and variant implementation

Aspect	Current Limitations	Emerging Prospects / Research Directions
Computational Efficiency	High training cost owing to dense collocation sampling and repeated PDE residual computation (Cuomo et al., 2022).	Domain decomposition (XPINN, cPINN), neural operators, and adaptive sampling were employed to enhance scalability and parallelization (Hu et al., 2021; Jagtap & Karniadakis, 2020; Zeinali et al., 2025).
Convergence Stability	Gradient imbalance between physics and data losses leads to slow or unstable convergence (Braga-Neto, 2021; McClenny & Braga-Neto, 2023).	Self-adaptive loss weighting, curriculum training, and residual-based adaptive refinement improve the convergence stability (Braga-Neto, 2021; Wang et al., 2025b).
Data Scarcity and Noise	Sparse or noisy environmental data limit the training accuracy (Waqas et al., 2024; Waqas et al., 2025).	Physics-informed priors, data assimilation (EnKF, 4D-Var), and data fusion PINNs enhance the robustness of the model under limited data (Hu et al., 2025).
Scalability in High Dimensions	Performance degrades in high-dimensional systems, such as 3D atmosphere–ocean coupling.	Fourier PINNs, Neural operators, and graph-based architectures offer scalable surrogate modeling (Mukkavilli et al., 2023)
Physical Consistency and Conservation	Standard PINNs may violate flux continuity and mass conservation in coupled domains.	The cPINN and vPINN enforce conservation principles through interface continuity (Jagtap & Karniadakis, 2020; Kharazmi et al., 2021; Kharazmi et al., 2019).
Uncertainty Quantification	Deterministic PINNs lack probabilistic interpretability.	sPINNs, Bayesian PINNs, and ensemble physics-informed models quantify uncertainty (Chen et al., 2021; Habib et al., 2025).
Integration with Operational Forecasting	Limited interoperability with legacy numerical models and data assimilation pipelines.	Development of hybrid frameworks and foundation physics-informed models for real-time (Chew et al., 2025; Mukkavilli et al., 2023).

5. Conclusion

This review highlights the transformative potential of PINNs in bridging the gap between data-driven learning and first-principles modeling in weather and hydrological modeling. PINNs provide a unified framework for learning non-linear and multi-scale processes, achieved through a single neural net that incorporates governing equations, thereby ensuring physical consistency. The vPINNs, XPINNs, cPINNs, sPINNs, fPINNs, and mesh-free PINNs address domain-related problems, including discontinuities, stochastic variability, anomalous diffusion, and irregular geometries. The combination of all these architectures introduces the flexibility of PINNs and makes them stronger when dealing with complex environmental systems.

Studies have demonstrated the application of PINNs in modeling the dynamics of convection-driven atmospheric flows, propagating flood waves, estimating groundwater properties, and synthesizing sparse or uncertain data. It is also worth noting that PINNs are not only surrogates for computationally expensive numerical solvers but also interpretable models that obey conservation laws. However, certain limitations are also evident, as seen in the review. PINNs require computationally intensive training, are unstable owing to stiff or high-dimensional problems, are sensitive to noise, and lack standardized frameworks for comparing results in hydrometeorological problems.

Recent developments, including adaptive sampling, neural operators, and hybrid systems that integrate PINNs with ensemble-based prediction and foundation models, have the potential to be scalable and viable solutions. Digital twins also consider PINNs a significant component of real-time observation of Earth systems, where precision, physical fidelity, and computational efficiency are essential.

PINNs can be considered as a family of models that are regularized by physics, and their usefulness varies based on the problem structure. They are suitable if the equations governing the system are reliable, observations are few, the parameters to be inverted are significant, or conservation constraints must be satisfied. They are less appealing in the case of dense labeled data, when physical constraints are less well understood, or when operational latency and robustness take precedence over physical interpretability. Reproducible benchmarking, field-scale validation, uncertainty quantification, and integration with current forecasting and data-assimilation infrastructure will become key to future progress.

Future research should focus on scalable architectures, the integration of PINN with numerical models, and uncertainty quantification methods to enhance reliability. The suggested improvements to hybrid frameworks, foundation models, and adaptive training strategies aim to develop physics-informed neural networks (PINNs) for real-time hydrometeorological forecasting.

List of Abbreviations

AI	Artificial Intelligence
ANN	Artificial Neural Network
BC	Boundary Condition
CNN	Convolutional Neural Network
cPINN	Conservative Physics-Informed Neural Network
DD-PINN	Domain-Decomposition Physics-Informed Neural Network
DL	Deep Learning
DNS	Direct Numerical Simulation
EnKF	Ensemble Kalman Filter
fPINNs	Fractional Physics-Informed Neural Network
GP	Gaussian Process
GPU	Graphics Processing Unit
HEC-RAS	Hydrologic Engineering Center – River Analysis System

IC	Initial Condition
KL	Karhunen–Loève (Expansion)
KLE-PINN	Karhunen–Loève Expansion PINN
L-BFGS	Limited-Memory Broyden–Fletcher–Goldfarb–Shanno
MAE	Mean Absolute Error
MCMC	Markov Chain Monte Carlo
MF-PINN	Mesh-Free Physics-Informed Neural Network
ML	Machine Learning
MLP	Multilayer Perceptron
MSE	Mean Squared Error
NWP	Numerical Weather Prediction
ODE	Ordinary Differential Equation
PDE	Partial Differential Equation
PC	Polynomial Chaos
PIML	Physics-Informed Machine Learning
PINN	Physics-Informed Neural Network
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
ReLU	Rectified Linear Unit
RNN	Recurrent Neural Network
RBF-PINN	Radial Basis Function PINN
RMSE	Root Mean Square Error
SA-PINN	Self-Adaptive Physics-Informed Neural Network
sPINNs	Stochastic Physics-Informed Neural Network
SWE	Shallow Water Equations
TD-PINN	Time-Difference Physics-Informed Neural Network
vPINN	Variational Physics-Informed Neural Network
VIC	Variable Infiltration Capacity Model
WRF	Weather Research and Forecasting Model
XPINN	Extended Physics-Informed Neural Network

759

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Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: